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中国开放指令生态（RISC-V）联盟
China RISC-V Alliance

香山开源处理器社区
XiangShan Open-Source Processor Community

XiangShan: An Open-Source High-Performance RISC-V Processor and Infrastructure for Architecture Research

The XiangShan Team

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What we will cover in this tutorial

- **Introduce of XiangShan (14:00 - 14:30, 30 minutes)**

- **CPU Microarchitecture (14:30 - 15:30, 60 minutes)**
  - *Design and implementation – How to implement novel ideas on XiangShan*
  - Frontend: branch prediction and instruction fetch
  - Backend: out-of-order scheduler, execution units
  - Load/Store Unit: LSQ, pipelines, TLBs, data caches
  - L2/L3 caches and prefetchers

- **Development workflows (16:00 - 17:30, 90 minutes)**
  - *Introduction of their usages – How to develop on XiangShan with MinJie*
  - Simulation and Debugging
  - Research Demo

In this tutorial, we will

- Provide access to cloud servers*
- Prepare the XiangShan development environment for you
- Go through the development workflows including:

Simulation

- Server login
- Environment
- RTL Generation
- RTL Simulation

……

Func. Verification

- Nexus-AM
- NEMU
- Difftest
- TL-Test

……

Perf. Verification

- Checkpoint
- Gem5
- Top-Down
- Constantin

……

- ***Required: machines with network and SSH***

Demo Instructions

- **Shell commands** are highlighted in **brown box** with prefix $

```
$ echo "Hello, XiangShan"
$ echo "Have a nice day"
```

- **Description and notes** are presented in **grey box** with prefix #

```
# Please prepare a laptop with an SSH client.
# Next, let's start the demo session !
```

Let's Start!

Prerequisites

- **Login to the provided cloud server**
  - **Windows (Windows Terminal / PowerShell is recommended)**

```
PS > ssh guest@t.xiangshan.cc
# Password: xiangshan-2025
```

  - **Linux / Mac**

```
$ ssh guest@t.xiangshan.cc
# Password: xiangshan-2025
$ guest@xiangshan-tutorial:~$ pwd
/home/guest
```

For **offline users**, please refer to https://github.com/OpenXiangShan/xs-env/tree/tutorial-2025

Prerequisites

```
# Copy tutorial environment to your dir based on your name
$ cp -r /opt/xs-env ~/<YOUR_NAME>

# Enter your dir
$ cd ~/<YOUR_NAME>

# Set up environment variables
# DO IT AGAIN when opening a new terminal
$ source env.sh

# SET XS_PROJECT_ROOT:                 /home/guest/YOUR_NAME
# SET NOOP_HOME (XiangShan RTL Home):  $XS_PROJECT_ROOT/XiangShan
# SET NEMU_HOME:                       $XS_PROJECT_ROOT/NEMU
# SET AM_HOME:                         $XS_PROJECT_ROOT/nexus-am
# SET TLT_HOME:                        $XS_PROJECT_ROOT/tl-test-new
# SET gem5_home:                       $XS_PROJECT_ROOT/gem5
```

Prerequisites

```
# Project Structure

$ tree -d -L 1

.
├── DRAMsim3
├── gem5
├── NEMU
├── nexus-am
├── NutShell
├── tl-test-new
├── tutorial
└── XiangShan

# Enter XiangShan directory

$ cd XiangShan
```

Chisel Compilation

NOT today, due to limited server resources

- **Compile RTL and build simulator with Verilator**

```
$ make emu -j4
Options:
    CONFIG=MinimalConfig          Configuration of XiangShan
    // EMU_THREADS=4              Simulation threads
    // EMU_TRACE=1                Enable waveform dump
    // WITH_DRAMSIM=1             Enable DRAMSim3 for DRAM simulation
    // WITH_CHISELDB = 1          Enable ChiselDB feature
    // WITH_CONSTANTIN = 1        Enable Constantin feature
```

- **Compilation might take ~20 mins**

Check your own emu

NOT today, due to limited server resources

- **After emu compilation**
- **Run simulation with your own emulator**

```
$ cd $NOOP_HOME
$ ./build/emu --help          (automatically runs with EMU_THREADS threads)
# Some Options:
    -i                        Workload to run
    -C / -I / -W              Max cycles / Max Insts / Warmup Insts
    --diff=PATH / --no-diff   Compare with NEMU for difftest / disable difftest
```

Example: Use the tab key for command completion

```
$ ./build/emu -i ../tutorial/p1-basic-func/hello.bin --diff ./ready-to-run/riscv64-nemu-interpreter-so 2> perf.err
```

Run RTL Simulation with Verilator

- **After building, we can run the simulator**

```
# run pre-built simulator
$ cd ../p1-basic-func
$ ./emu -i hello.bin --no-diff 2>hello.err

# Some key options:
    -i                         # Workload to run
    -C / -I                    # Max cycles  /Max Insts
    --diff=PATH / --no-diff    # Path of Reference Model / disable difftest
```

Great!

We have learned the basic simulation process of Xiangshan.

Once we get the RTL simulator ready, what's next?

How can we

- **generate desired workloads**
- **find and fix functional bugs**
- **do performance analysis**
- **do research on micro-architecture**

MinJie Functional Verification

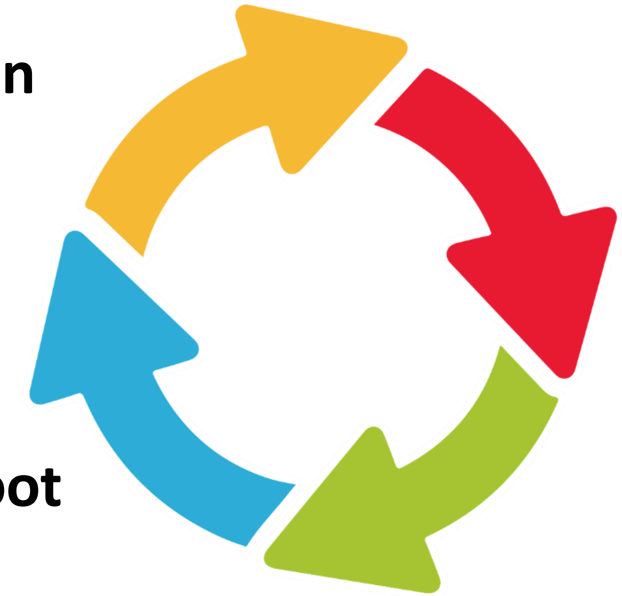
Nexus-AM

Test Generation

Bug Detection

Difftest

NEMU



Waveform

ChiselDB

Troubleshoot & Fix bugs

Context Preservation

LightSSS

Nexus-AM: Bare Metal Runtime

- **Purpose**
  - Generate workloads **agilely** without an OS
  - Provide runtime framework for **bare metal machines** like XiangShan
- **We present a bare metal runtime called Nexus-AM**
  - Light-weight, easy to use
  - Implement basic **system call** interfaces and exception handlers
  - Support multiple **ISAs and configurations**

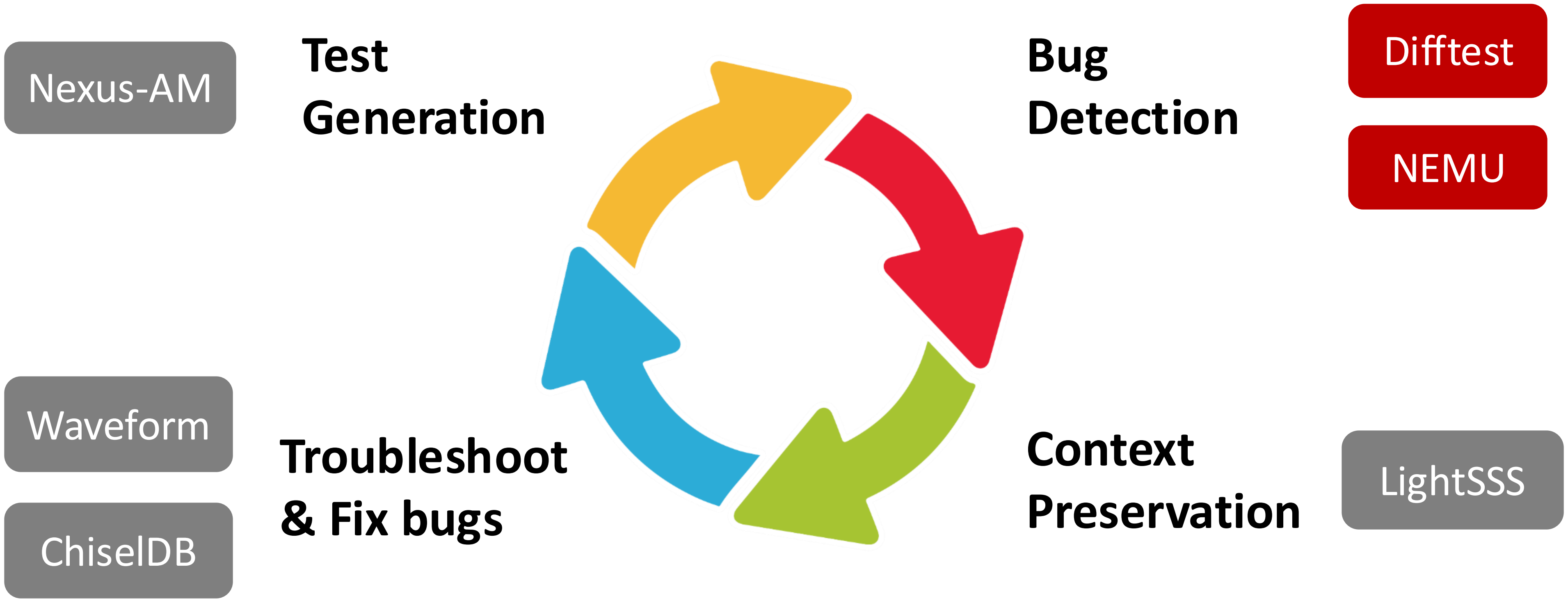
Nexus-AM

- **Hands-on:** build Coremark workload
- **Showcase**

```
$ bash build-am.sh
```

```
# cd $AM_HOME/apps/coremark
# make ARCH=riscv64-xs \
  ITERATIONS=1 LINUX_GNU_TOOLCHAIN=1 TOTAL_DATA_SIZE=400
# ls -l build
#     coremark-riscv64-xs.bin    Binary image of the program
#     coremark-riscv64-xs.elf    ELF of the program
#     coremark-riscv64-xs.txt    Disassembly of the program
```

MinJie Functional Verification



NEMU: ISA REF

- **Purpose**
  - Golden model for verification
  - Simple yet high performance
- **We present NEMU**
  - An **instruction set simulator**, similar to Spike
  - Through **optimization techniques**, achieve performance similar to QEMU.
  - **A set of APIs** is provided to assist XiangShan in comparing and verifying the **architectural states**

NEMU

- **Showcase**

```
$ bash build-nemu.sh
```

```
# cd $NEMU_HOME
# make clean
# make riscv64-xs_defconfig
# make -j      build NEMU as the bare metal machine, can run the Coremark from the previous step
# make clean-softfloat
# make riscv64-xs-ref_defconfig
# make -j      build NEMU as the reference model for XiangShan
```

NEMU

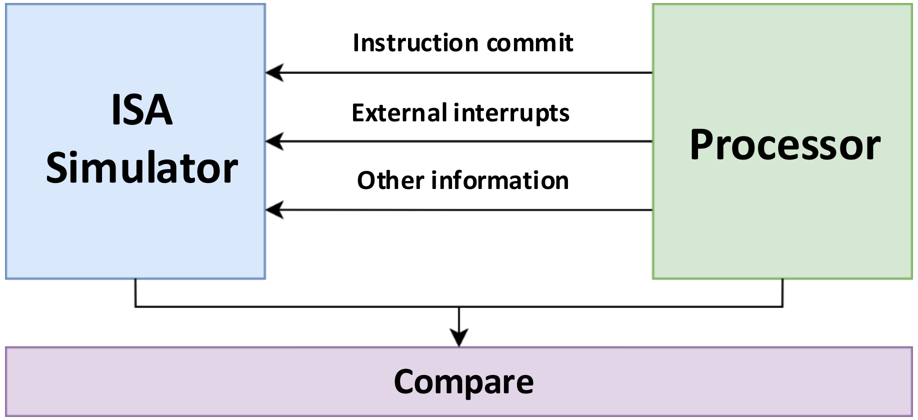
- **Hands-on:** run Coremark workload on NEMU
- **Showcase**

```
$ bash run-nemu.sh
```

```
# cd $NEMU_HOME                                   run in batch mode, faster
# ./build/riscv64-nemu-interpreter -b \           set workspace to Coremark
      $AM_HOME/apps/coremark/build/coremark-1-iteration-riscv64-xs.bin
```

Difftest: ISA Co-simulation Framework

- **Basic flows**
  - Instructions commit/other states update
  - The simulator executes the same instructions
  - Compare the architectural states between DUT and REF
  - Abort or continue
- **Provided as verification infrastructure for processors**
  - APIs for HDLs such as Chisel/Verilog
  - RTL simulators such as Verilator, VCS
  - RISC-V Instruction Set Simulators, such as **NEMU**, **Spike**
- **SMP-Difftest: co-simulation on an SMP processor**
  - SMP Linux kernel and multi-threading programs
  - Online checking of cache coherency and memory consistency



Basic architecture of DiffTest

```
while (1) {
    icnt = cpu_step();
    nemu_step(icnt);
    r1s = cpu _getregs();
    r2s = nemu_getregs();
    if (r1s != r2s) { abort();
}
}
```

Online checking

Difftest

NOT today, due to limited server resources

- **Hands-on:** run Coremark workload on XiangShan, difftest with NEMU

- **Showcase**

```
# Running Coremark takes about 3 minutes
$ bash run-emu.sh
```

```
# ./emu -i \    use coremark.bin
$AM_HOME/apps/coremark/build/coremark-1-iteration-riscv64-xs.bin
--diff $NEMU_HOME/build/riscv64-nemu-interpreter-so \      difftest with NEMU
2>perf.out   \     redirect stderr to file
```

Difftest

```
./emu -i $AM_HOME/apps/coremark/build/coremark-1-iteration-riscv64-xs.bin --diff $NEMU_HOME/build/riscv64-nemu-interpreter-so 2>perf.err
emu compiled at Feb 21 2025, 10:49:46
Using simulated 32768B flash
Using simulated 8386560MB RAM
The image is /nfs/home/mayuexiao/tutorial/xs-env/nexus-am/apps/coremark/build/coremark-1-iteration-riscv64-xs.bin
The reference model is /nfs/home/mayuexiao/tutorial/xs-env/NEMU/build/riscv64-nemu-interpreter-so
The first instruction of core 0 has commited. Difftest enabled.
Running CoreMark for 1 iterations
2K performance run parameters for coremark.
CoreMark Size     : 666
Total time (ms)   : 1931
Iterations        : 1
Compiler version  : GCC11.4.0
seedcrc           : 0xe9f5
[0]crclist        : 0xe714
[0]crcmatrix      : 0x1fd7
[0]crcstate       : 0x8e3a
[0]crcfinal       : 0xe714
Finished in 1931 ms.
==================================================
CoreMark Iterations/Sec 517.87
Core 0: HIT GOOD TRAP at pc = 0x80001de2
Core-0 instrCnt = 379,235, cycleCnt = 252,495, IPC = 1.501951
Seed=0 Guest cycle spent: 252,499 (this will be different from cycleCnt if emu loads a snapshot)
Host time spent: 164,455ms
```

Difftest

- **Hands-on:**
  - We injected a bug in a pre-built XiangShan processor
  - Now, let's trigger it by Difftest
- **Showcase**

```
$ bash run-emu-diff.sh
```

```
# ./emu-bug \
  -i $AM_HOME/apps/coremark/build/coremark-1-teration-riscv64-xs.bin \
  --diff $NEMU_HOME/build/riscv64-nemu-interpreter-so    # difftest with NEMU
```

Difftest

- Difftest will report the error once failed
  - Dump info like PC, GPRs, etc.

```
============== Commit Group Trace (Core 0) ============
commit group [00]: pc 0080001654 cmtcnt 9
commit group [01]: pc 008000166a cmtcnt 6
commit group [02]: pc 0080001a6e cmtcnt 4
commit group [03]: pc 0080001a7a cmtcnt 1
commit group [04]: pc 0080001a7c cmtcnt 2
commit group [05]: pc 008000167a cmtcnt 2
commit group [06]: pc 008000166c cmtcnt 6 <--
commit group [07]: pc 00800015f6 cmtcnt 1
commit group [08]: pc 00800015f8 cmtcnt 2
commit group [09]: pc 00800015fc cmtcnt 1
commit group [10]: pc 0080001600 cmtcnt 1
commit group [11]: pc 008000160e cmtcnt 3
commit group [12]: pc 0080001614 cmtcnt 1
commit group [13]: pc 008000162a cmtcnt 1
commit group [14]: pc 008000162e cmtcnt 4
commit group [15]: pc 008000163e cmtcnt 8
```

```
==============  REF Regs  ==============
---------------- Intger Registers ----------------
  $0: 0x0000000000000000   ra: 0x00000000800015f0   sp: 0x000000008000cef0   gp: 0x0000000000000000
  tp: 0x0000000000000000   t0: 0x0000000000000000   t1: 0x0000000000000001   t2: 0x0000000000000009
  s0: 0x0000000000000000   s1: 0x0000000000000000   a0: 0x000000000000029a   a1: 0x000000008000cf08
  a2: 0x0000000000000002   a3: 0x000000000000029a   a4: 0x0000000000000002   a5: 0x0000000000000007
  a6: 0x0000000000000001   a7: 0x0000000000000003   s2: 0x0000000000000000   s3: 0x0000000000000000
  s4: 0x0000000000000000   s5: 0x0000000000000000   s6: 0x0000000000000000   s7: 0x0000000000000000
  s8: 0x0000000000000000   s9: 0x0000000000000000  s10: 0x0000000000000000  s11: 0x0000000000000000
  t3: 0x00000000800039b8   t4: 0x0000000000000001   t5: 0x000000008000cd91   t6: 0x0000000000000031
---------------- Float Registers ----------------
 ft0: 0xffffffff00000000  ft1: 0xffffffff00000000  ft2: 0xffffffff00000000  ft3: 0xffffffff00000000
 ft4: 0xffffffff00000000  ft5: 0xffffffff00000000  ft6: 0xffffffff00000000  ft7: 0xffffffff00000000
 fs0: 0xffffffff00000000  fs1: 0xffffffff00000000  fa0: 0xffffffff00000000  fa1: 0xffffffff00000000
 fa2: 0xffffffff00000000  fa3: 0xffffffff00000000  fa4: 0xffffffff00000000  fa5: 0xffffffff00000000
 fa6: 0xffffffff00000000  fa7: 0xffffffff00000000  fs2: 0xffffffff00000000  fs3: 0xffffffff00000000
 fs4: 0xffffffff00000000  fs5: 0xffffffff00000000  fs6: 0xffffffff00000000  fs7: 0xffffffff00000000
 fs8: 0xffffffff00000000  fs9: 0xffffffff00000000 fs10: 0xffffffff00000000 fs11: 0xffffffff00000000
 ft8: 0xffffffff00000000  ft9: 0xffffffff00000000 ft10: 0xffffffff00000000 ft11: 0xffffffff00000000
 fcsr: 0x0000000000000000 fflags: 0x0000000000000000 frm: 0x0000000000000000
```

```
privilegeMode: 3
     a1 different at pc = 0x008000166c, right= 0x000000008000cf08, wrong = 0x000000008000cf10
     a3 different at pc = 0x008000166c, right= 0x000000000000029a, wrong = 0x0000000000000000
Core 0: ABORT at pc = 0x80001694
Core-0 instrCnt = 1,264, cycleCnt = 10,610, IPC = 0.119133
Seed=0 Guest cycle spent: 10,613 (this will be different from cycleCnt if emu loads a snapshot)
Host time spent: 27,813ms
```

Difftest

- **Hands-on:** Dump waveform after Difftest detects the bug

- **Showcase**

```
$ bash run-emu-dumpwave.sh
$ ls $NOOP_HOME/build/*.vcd -lh    # There should be a .vcd waveform
```

```
# ./emu-bug \
-i $AM_HOME/apps/coremark/build/coremark-1-iteration-riscv64-xs.bin \
--diff $NEMU_HOME/build/riscv64-nemu-interpreter-so \
-b 9000 \         the begin cycle of the wave, you may find XiangShan trap at 10000 cycles in previous step
-e 11000 \        the end cycle of the wave
--dump-wave \     set this option to dump wave, you will find *.vcd on $NOOP_HOME/build
```

MinJie Functional Verification

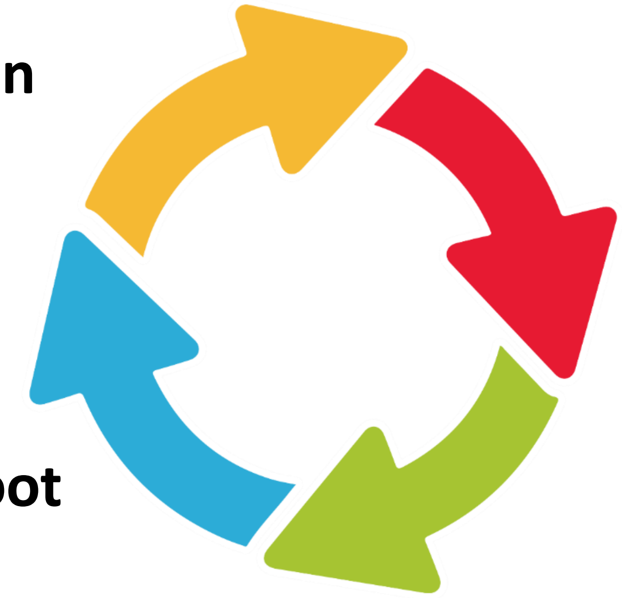
Nexus-AM

Test Generation

Bug Detection

Difftest

NEMU



Waveform

ChiselDB

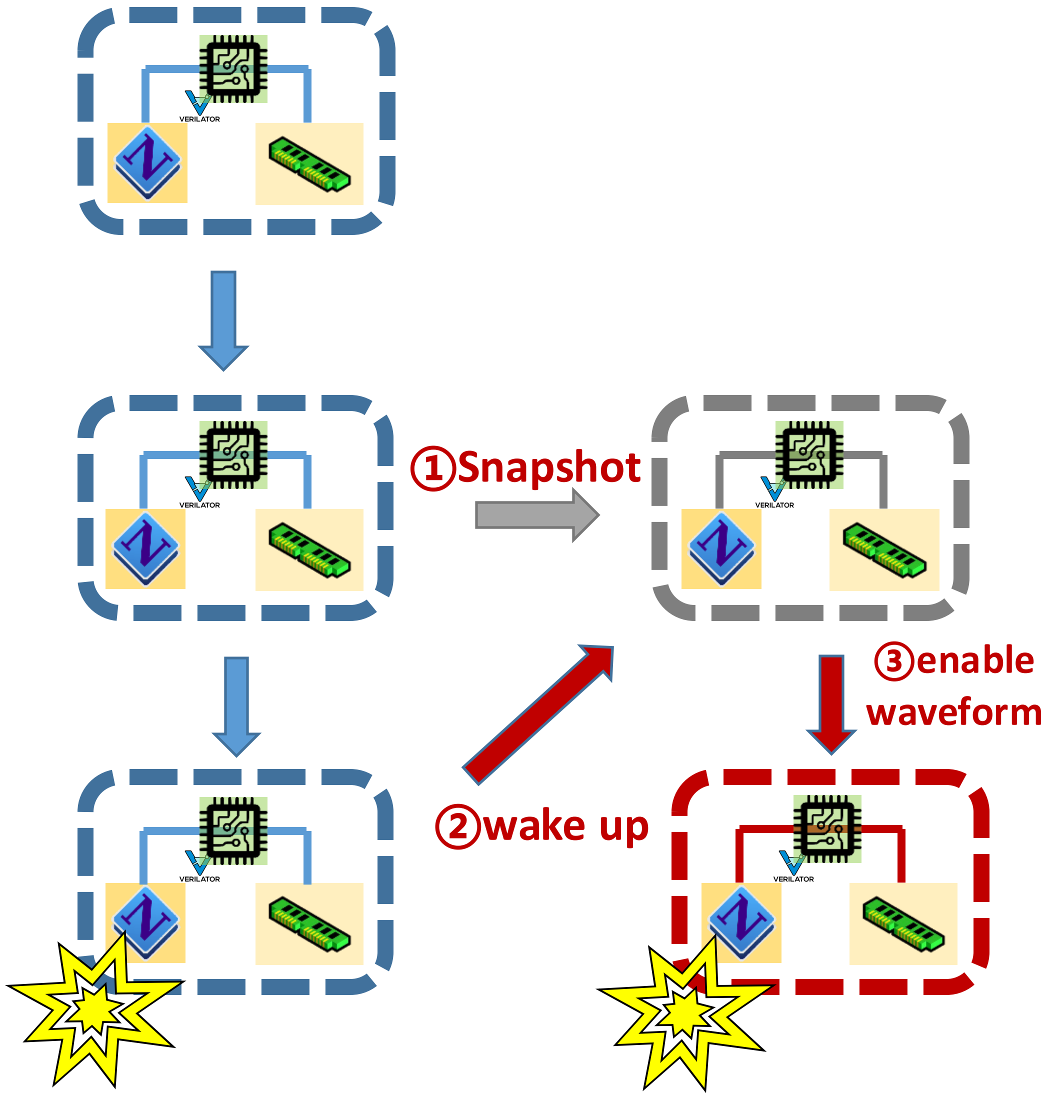
Troubleshoot & Fix bugs

Context Preservation

LightSSS

LightSSS: Light-weight Simulation SnapShot

- **Re-run simulation is time-consuming!**
  - **Snapshot is the way out**
- **LightSSS: snapshots of the process with fork()**
  - Copy-on-write on Linux Kernel
- **Advantage 1: Good portability and scalability**
  - Snapshots for any external models (such as C++)
  - No need to understand details of external models
- **Advantage 2: Low overhead of snapshots**
  - ~500us for taking a snapshot
  - Far less overhead than RTL snapshots by Verilator



LightSSS

- **Hands-on:** use lightSSS to dump waveform
- **Showcase**

```
$ bash run-emu-lightsss.sh
```

```
# ./emu-bug \
-i $AM_HOME/apps/coremark/build/coremark-1-iteration-riscv64-xs.bin \
--diff $NEMU_HOME/build/riscv64-nemu-interpreter-so \
--enable-fork \      No longer need to use complex parameters
2 > lightsss.err
```

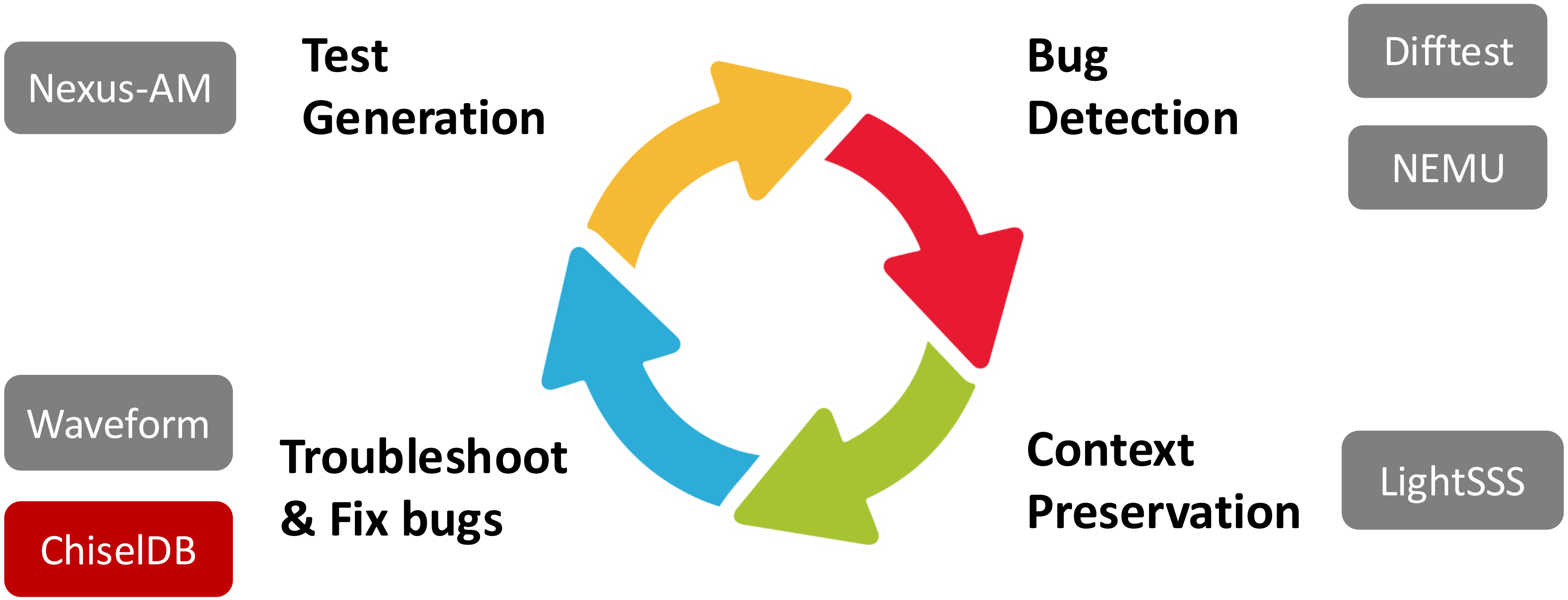
LightSSS

- LightSSS is working if you see:

```
privilegeMode: 3
     a1 different at pc = 0x008000166c, right= 0x000000008000cf08, wrong = 0x000000008000cf10
     a3 different at pc = 0x008000166c, right= 0x000000000000029a, wrong = 0x0000000000000000
[FORK_INFO pid(1459543)] the oldest checkpoint start to dump wave and dump nemu log...
[FORK_INFO pid(1459543)] dump wave to /nfs/home/mayuexiao/tutorial/xs-env/XiangShan/build/2025-02-21@17:09:13_1.vcd...
%Warning: previous dump at t=0, requesting t=0, dump call ignored
The first instruction of core 0 has commited. Difftest enabled.
Running CoreMark for 1 iterations

==============  In the last commit group  ==============
the first commit instr pc of DUT is 0x000000008000166c
the first commit instr pc of REF is 0x000000008000166c

============== Commit Group Trace (Core 0) ==============
commit group [00]: pc 0080001654 cmtcnt 9
commit group [01]: pc 008000166a cmtcnt 6
commit group [02]: pc 0080001a6e cmtcnt 4
```

- After that, simulation will start again from the existing latest snapshot

MinJie Functional Verification



ChiselDB: Debug-friendly Structured Database

- **Motivation:**
  - Waveforms are **large** in size and **hard to apply** further analysis
  - Need to analyze structured data like memory transaction trace
- **We present ChiselDB**
- **Highlights:**
  - Inserting probes between module interfaces in hardware
  - DPI-C: Using C++ function in Chisel code to transfer data
  - Persist in database, SQL queries supported

ChiselDB

- **Design source code:** *XiangShan/utility/src/main/scala/utility/ChiselDB.scala*
- **Usage:** Create table

```
// API: def createTable[T <: Record](tableName: String, hw: T): Table[T]
import huancun.utils.ChiselDB

class MyBundle extends Bundle {
        val fieldA = UInt(10.W)
        val fieldB = UInt(20.W)
}
val table = ChiselDB.createTable("MyTableName", new MyBundle)
```

ChiselDB

- **Usage:** Register probes

```
/* APIs
def log(data: T, en: Bool, site: String = "", clock: Clock, reset: Reset)
def log(data: Valid[T], site: String, clock: Clock, reset: Reset): Unit
def log(data: DecoupledIO[T], site: String, clock: Clock, reset: Reset): Unit
*/
table.log(
  data = my_data /* hardware of type T */,
  en = my_cond,    site = "MyCallSite",
  clock = clock,    reset = reset
)
```

ChiselDB

- **Example:** *XiangShan/src/main/scala/xiangshan/cache/mmu/L2TLB.scala*

```scala
class L2TlbMissQueueDB(implicit p: Parameters) extends TlbBundle {
   val vpn = UInt(vpnLen.W)
}

val L2TlbMissQueueInDB, L2TlbMissQueueOutDB = Wire(new L2TlbMissQueueDB)
L2TlbMissQueueInDB.vpn := missQueue.io.in.bits.vpn
L2TlbMissQueueOutDB.vpn := missQueue.io.out.bits.vpn

val L2TlbMissQueueTable = ChiselDB.createTable(
        "L2TlbMissQueue_hart" + p(XSCoreParamsKey).HartId.toString, new L2TlbMissQueueDB)

L2TlbMissQueueTable.log(L2TlbMissQueueInDB, missQueue.io.in.fire, "L2TlbMissQueueIn", clock, reset)
L2TlbMissQueueTable.log(L2TlbMissQueueOutDB, missQueue.io.out.fire, "L2TlbMissQueueOut", clock, reset)
```

ChiselDB

- **TL-Log： Persistence Transactions/Database**
  - Record bus transactions to SQLite3 database using ChiselDB
  - Support offline bug detect and performance analysis

Time	Name	Channel	Opcode	Permission	State	SourceID	SinkID	Address	Data
116854134	L2_L1I_0	A	AcquireBlock	Grow	NtoB	1	0	803d3680	0000000000000000 0000000000000000 0000000000000000 0000000000000000
116854146	L2_L1I_0	D	GrantData	Cap	toT	1	16	803d3680	fa8437839043903 06090063843c23 84382300093783 378300093023c3bd
116854147	L2_L1I_0	D	GrantData	Cap	toT	1	16	803d3680	03e32e07bb030089 f0ef8526c881f49b b60300893783bbdf 855a010925832e07
123191840	L2_L1I_0	C	ReleaseData	Shrink	TtON	0	0	803d3680	fa8437839043903 06090063843c23 84382300093783 378300093023c3bd
123191841	L2_L1I_0	B	Probe	Cap	toN	0	0	803d3680	0000000000000000 0000000000000000 0000000000000000 0000000000000000
123191841	L2_L1I_0	C	ReleaseData	Shrink	TtoN	0	0	803d3680	03e32e07bb030089 f0ef8526c881f49b b60300893783bbdf 855a010925832e07
123191848	L2_L1I_0	D	ReleaseAck	Cap	toT	0	31	803d3680	0000000020ab05b 0000000020fab45b 0000000020fab85b 0000000020fabc5b
123191851	L3_L2_0	C	ReleaseData	Shrink	TtoN	31	0	803d3680	fa8437839043903 06090063843c23 84382300093783 378300093023c3bd
123191852	L3_L2_0	C	ReleaseData	Shrink	TtoN	31	0	803d3680	03e32e07bb030089 f0ef8526c881f49b b60300893783bbdf 855a010925832e07
123191862	L3_L2_0	D	ReleaseAck	Cap	toT	31	16	803d3680	86a6f7043583dcd5 f0ef856a865a8762 3583b7654481e17 865a86a68762f704
123191863	L2_L1I_0	C	ProbeAck	Shrink	TtoN	0	0	803d3680	0000000000000000 0000000000000000 0000000000000000 0000000000000000
123191878	L3_L2_0	C	Release	Report	NtoN	30	0	803d3680	5597bf494981aa09 8522cc2585930000 f84ef15dda0f00ef 001947036775e84a

Example: requests sequence in database that violate cache coherence

ChiselDB

- **Hands-on: Analysis of Cache Coherence Violation**

```
# Inject bug – We set all Release Data (L2->L3) as a constant value
$ cd ../p2-chiseldb && cat cdb_err.patch
```

```
--- a/src/main/scala/huancun/noninclusive/SinkC.scala
+++ b/src/main/scala/huancun/noninclusive/SinkC.scala

@@ -99,7 +99,7 @@ class SinkC(implicit p: Parameters) extends
BaseSinkC {
-       buffer(insertIdx)(count) := c.bits.data
+       buffer(insertIdx)(count) := 0xABCDEF.U
```

ChiselDB

- **Hands-on: Analysis of Cache Coherence Violation**

```
# simulate using pre-built emu (ChiselDB enabled)
$ bash chiseldb_step1_run.sh
```

```
# ./emu-cdb-err -i $NOOP_HOME/ready-to-run/linux.bin \
  --diff $NOOP_HOME/ready-to-run/riscv64-nemu-interpreter-so \
  --dump-db     # set this argument to dump data base
  2>linux.err
```

ChiselDB

- **Hands-on: Analysis of Cache Coherence Violation**

- **Difftest Info**
  - Error Address： 0x80040580

```
The first instruction of core 0 has commited. Difftest enabled.
cacheid=1,idtfr=0,realpaddr=0x80040580: Refill test failed!
addr: 80040580
Gold: 000000008000447e00000000800044aa000000008000ce32000000008000ce1000000000800043f40000000080004412000000008004190000000000800419e0
Core: 0000000000abcdef0000000000000000000000000000000000000000000000000000000000abcdef000000000000000000000000000000000000000000000000
Core 0: ABORT at pc = 0x8000474e
Core-0 instrCnt = 9,128, cycleCnt = 12,539, IPC = 0.727969
```

ChiselDB

- **Hands-on: Analysis of Cache Coherence Violation**

```
# Analyze
$ bash chiseldb_step2_analyze.sh
```

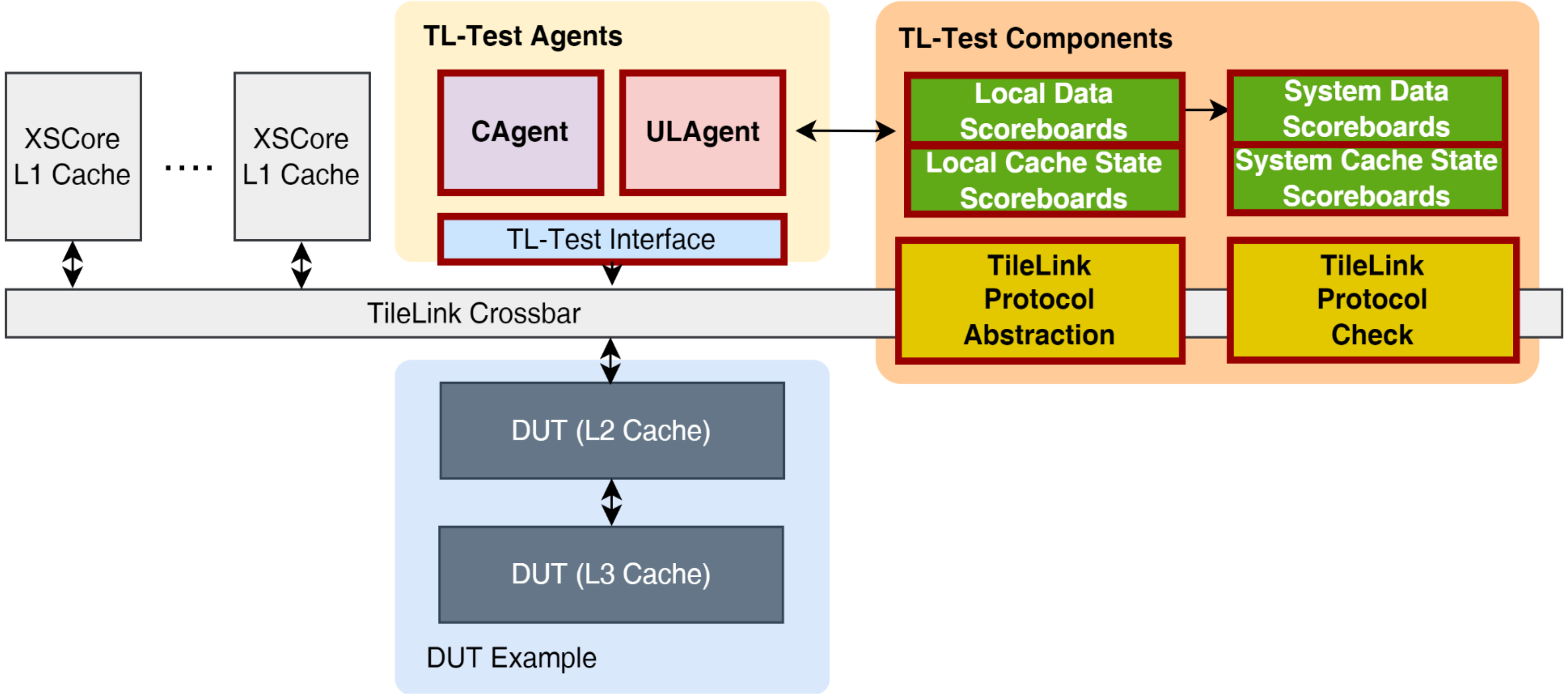
```
# 1. sqlite query for all transactions on Eaddr
# 2. use script to parse TLLog
# sqlite3 $NOOP_HOME/build/*.db \
  "select * from TLLOG where ADDRESS=0x80040580" | \
  sh $NOOP_HOME/scripts/utils/parseTLLog.sh
```

ChiselDB:

```
# Result: [Time | To_From | Channel | Opcode | Permission | Address | Data]

# Data successfully transferred from L1D to L2
11116 L2_L1D_0 C ReleaseData Shrink TtoN 80040580
000000000000447e 00000000000044aa 000000008000ce32 000000008000ce10
# Data successfully transferred from L2 to L3
12081 L3_L2_0 C ProbeAckData Shrink TtoN 80040580
000000000000447e 00000000000044aa 000000008000ce32 000000008000ce10


# But when L1D acquires Eaddr again, data loaded from L3 is wrong
12492 L2_L1D_0 A AcquireBlock Grow NtoB 80040580
12498 L3_L2_0 A AcquireBlock Grow NtoB 0 1 80040580
12511 L3_L2_0 D GrantData Cap toT 80040580
0000000000abcdef 0000000000000000 0000000000000000 0000000000000000
# So there must be something wrong when L3 records Release Data
```

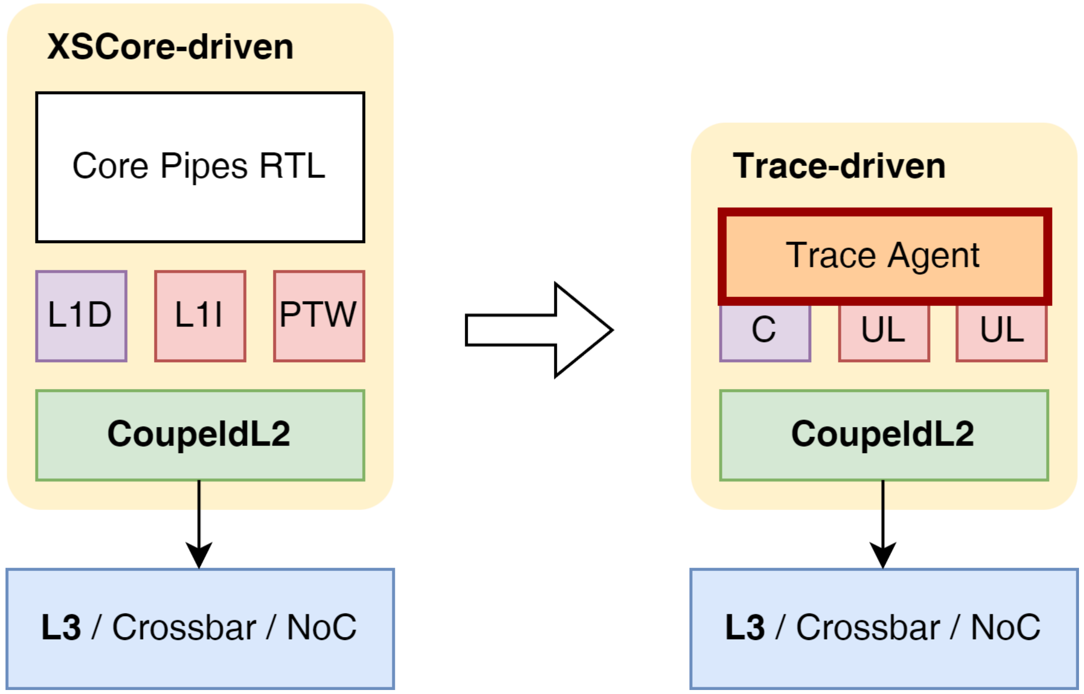
TL-Test: Cache Verification Framework

- **As an infrastructure for cache verification**
  - Support **Tilelink protocol** and **cache coherence** check
  - **Fast test cases generation** and **constrained random test**



TL-Test

- **TL-Trace: Trace-based testbench generation**
  - Convert cache access trace into testbench
  - Fast reproduction of functional bugs and analysis of performance problems



TL-Test

 # Project Structure

```
$ cd $TLT_HOME && tree -d -L 1
.
├── assets
├── configs
├── dut
├── main
├── run
└── scripts
```

 # Enter TL-Test tutorial

```
$ cd ../tutorial/p3-tl-test
```

TL-Test

- **Hands-on: Analysis of Cache Coherence Violation**

```
# Inject bug – We wrongly shift the Grant Data (L2->L1)
$ cat tlt_err.patch
```

```
--- a/tl-test-new/dut/CoupledL2/src/main/scala/coupledL2/GrantBuffer.scala
+++ b/tl-test-new/dut/CoupledL2/src/main/scala/coupledL2/GrantBuffer.scala
@@ -94,7 +94,7 @@ class GrantBuffer(implicit p: Parameters) extends
LazyModule {
-       d.data := data
+       d.data := data << 8
```

TL-Test

- **Hands-on: Analysis of Cache Coherence Violation**

```
# We provide a pre-compiled TL-Test

# Run the TL-Test

$ bash tltest_step1_run.sh
```

```
# cd $TLT_HOME/run && ./tltest_v3lt 2>&1 | tee tltest_v3lt.log
```

TL-Test

- **Hands-on: Analysis of Cache Coherence Violation**

- TL-Test Info
  - Error Address： 0x4000

```
[2940] [tl-test-new-INFO] #0 L2[0].C[0] [data complete D] [GrantData toT] source: 0, addr: 0x4000, alias: 0, data: [ 00 ff ... 00 70 ... ]

dut: [ 00 ff ... 00 70 ... ]
ref: [ ff e6 ... 70 f9 ... ]
[2940] [tl-test-new-ERROR] [tlc_assert failure at int GlobalBoard<unsigned long>::data_check(TLLocalContext *, const uint8_t *, const uint8_t *,
 std::string) [T = unsigned long]:277]
[2940] [tl-test-new-ERROR] [tlc_assert failure from system #0]
[2940] [tl-test-new-ERROR] info: Data mismatch from status SB_VALID!
```

TL-Test

- **Hands-on: Analysis of Cache Coherence Violation**

```
# Analyze
$ bash tltest_step2_analyze.sh
```

```
# 1. grep all transactions on Eaddr(0x4000)
# 2. use TL-Log to help debugging

# grep "addr: 0x4000" $TLT_HOME/run/tltest_v3lt.log
```

TL-Test

```
# Result: [Time | Core | Channel | Opcode | Source | Address | Data]

# L1D acquires Eaddr
[1954] L2[0].C[0] [fire A] [AcquireBlock NtoB] source: 0, addr: 0x4000

# L2 probes L1D, data successfully transferred from L1D to L2
[2190] L2[0].C[0] [fire B] [ProbeBlock toN] source: 0, addr: 0x4000
[2192] L2[0].C[0] [fire C] [ProbeAckData TtoN] source: 0x10, addr: 0x4000, data: [ ff e6 ... ]
[2194] L2[0].C[0] [fire C] [ProbeAckData TtoN] source: 0x10, addr: 0x4000, data: [ 70 f9 ... ]

# But when L2 grants data of Eaddr, data loaded from L2 is wrong
[2938] L2[0].C[0] [fire D] [GrantData toT] source: 0x1, addr: 0x16000, data: [ 00 ff... ]
[2940] L2[0].C[0] [fire D] [GrantData toT] source: 0x1, addr: 0x16000, data: [ 00 70... ]

# So there must be something wrong when L2 Grant Data
```

MinJie Performance Verification

Chisel-based prototypes

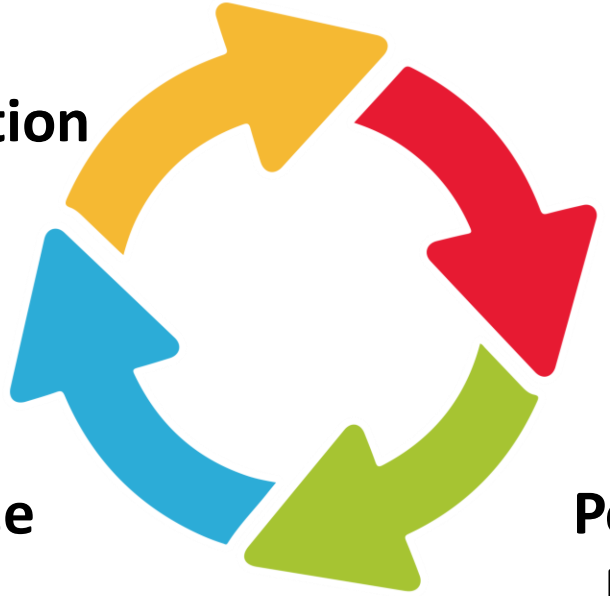
Implementation

Run Tests

gem5

VERILATOR

umd-memsys/
DRAMsim3
DRAMsim3: a Cycle-accurate, Thermal-Capable DRAM Simulator



XSPerf

Constantin

Top-Down

Performance Analysis

Performance Evaluation

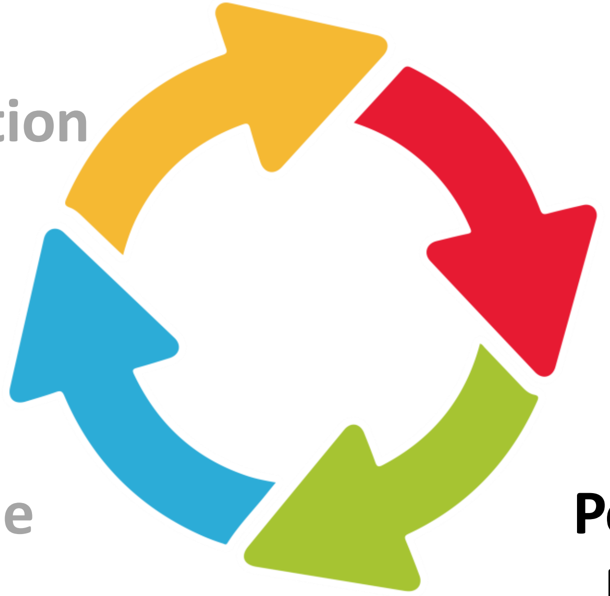
RISC-V Checkpoint

SimPoint

MinJie Performance Verification

Chisel-based prototypes

Implementation



Run Tests

gem5

VERILATOR

umd-memsys/
DRAMsim3

DRAMsim3: a Cycle-accurate, Thermal-Capable DRAM Simulator

XSPerf

Constantin

Top-Down

Performance Analysis

Performance Evaluation

RISC-V Checkpoint

SimPoint

Checkpoint: Agile Performance Evaluation

High-performance designs are too complex to develop agilely on FPGA

Large

Design Size

RTL-Simulation

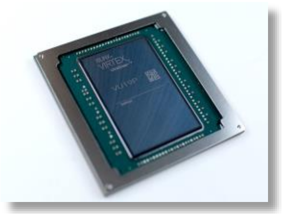
VERILATOR

Emulator: Million $
(Unaffordable for us and most academia)

Slow

Speed

Fast



FPGA

How to speed up the simulation?

Small

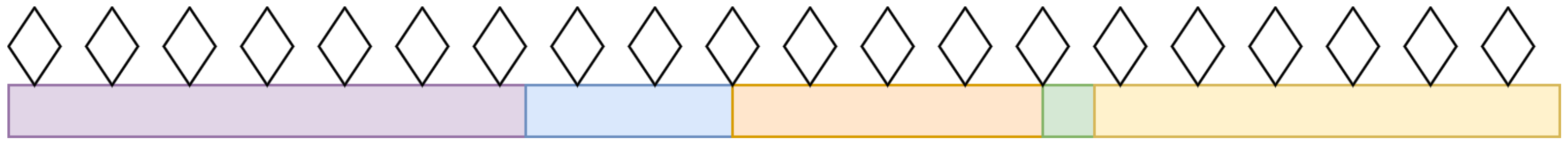
Time for performance modelling of single-core XiangShan on SPEC CPU2006

	RTL-simulation	FPGA	RTL-Simulation w/ Checkpoint
Compile	20 minutes	5 hours	20 minutes
Simulation	958 years@2KHz (2K CPS)	7 days@100MHz	5.5 hours with enough x86 servers

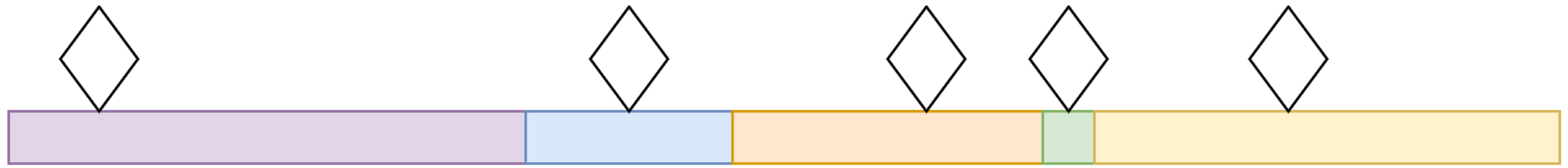
Our Approach

Checkpoint: Agile Performance Evaluation

- Uniform Checkpoint
  - Divide a program to equal segments for parallel simulation with uniform sampling



- SimPoint Checkpoint
  - Select representative segments with cluster analysis and weighted sampling



Checkpoint: SimPoint Checkpoint

- **Step 0: Prepare NEMU environment for SimPoint checkpoint**

```
$ cd ../p4-checkpoint && bash simpoint_step0_prepare.sh
```

```
# simpoint_step0_prepare.sh (~20s)

# cd $NEMU_HOME
# git submodule update --init
# cd $NEMU_HOME/resource/simpoint/simpoint_repo
# make clean && make                     # generate simpoint generator binary

# cd $NEMU_HOME
# make clean
# make riscv64-xs-cpt_defconfig && make -j 8      # compile NEMU

# cd $NEMU_HOME/resource/gcpt_restore
# rm -rf $XS_PROJECT_ROOT/tutorial/part5-checkpoint/gcpt
# make -C $NEMU_HOME/resource/gcpt_restore/ \      # generate gcpt restorer binary
        O=$XS_PROJECT_ROOT/tutorial/part5-checkpoint/gcpt \ # directory of results
        GCPT_PAYLOAD_PATH=$XS_PROJECT_ROOT/tutorial/part5-
        checkpoint/bin/stream_100000.bin
```

Checkpoint: SimPoint Checkpoint

- **Step 1: Execute the workload and collect program behavior**

```
$ bash simpoint_step1_profiling.sh
```

```
# simpoint_step1_profiling.sh (~20s)
# source simpoint_env.sh                    # configure environment variables

# rm -rf $RESULT

# $NEMU ${BBL_PATH}/${workload}.bin \       # specify workload
#     -b                            \       # run with batch mode
#     -D $RESULT                    \       # directory where the checkpoint is generated
#     -C $profiling_result_name     \       # name of this task
#     -w stream                     \       # name of workload
#     --simpoint-profile            \       # is simpoint profiling
#     --cpt-interval ${interval}    \       # simpoint interval instructions: 50,000,000
#     > >(tee $log/stream-out.txt) 2> >(tee ${log}/stream-err.txt) # redirect stdout and stderr
```

Checkpoint: SimPoint Checkpoint

```
-------------------------------------------------------------
Your clock granularity/precision appears to be 26 microseconds.
Each test below will take on the order of 1023153 microseconds.
   (= 39352 clock ticks)
Increase the size of the arrays if this shows that
you are not getting at least 20 clock ticks per test.
-------------------------------------------------------------
WARNING -- The above is only a rough guideline.
For best results, please be sure you know the
precision of your system timer.
-------------------------------------------------------------
Function    Best Rate MB/s  Avg time     Min time     Max time
Copy:           941.2     0.085167     0.084997     0.085384
Scale:          215.4     0.371746     0.371488     0.371877
Add:            275.9     0.435193     0.434995     0.435358
Triad:          256.4     0.468193     0.467980     0.468407
-------------------------------------------------------------
Solution Validates: avg error less than 1.000000e-06 on all three arrays
-------------------------------------------------------------
[src/monitor/monitor.c:195,parse_args] Doing Simpoint Profiling
[src/checkpoint/path_manager.cpp:54,init] Cpt id: -1
[src/checkpoint/path_manager.cpp:72,setSimpointProfilingOutputDir] Created /home/guest/fenghaoyuan/tutorial/p4-checkpoint/simpoint_result/simpoint-profiling/stream/

[src/checkpoint/simpoint.cpp:83,init] Doing simpoint profiling with interval 50000000
[src/checkpoint/serializer.cpp:378,next_index] set next index 0
[src/checkpoint/path_manager.cpp:91,setCheckpointingOutputDir] donot set checkpoint path without Checkpoint mode
[src/isa/riscv64/init.c:191,init_isa] NEMU will start from pc 0x80000000
[src/monitor/image_loader.c:203,load_img] Loading image (checkpoint/bare metal app/bbl) form cmdline: /home/guest/fenghaoyuan/tutorial/p4-checkpoint/gcpt/build/gcpt.
```

NEMU will do profiling while simulating workload STREAM at intervals of 50,000,000 instructions.

Checkpoint: SimPoint Checkpoint

- **Step 2: Cluster and obtain multiple representative slices**

```
$ bash simpoint_step2_cluster.sh
```

```
# simpoint_step2_cluster.sh

# export CLUSTER=$RESULT/cluster/${workload} && mkdir -p $CLUSTER
# mkdir -p $LOG_PATH/cluster_logs/cluster

# random1=`head -20 /dev/urandom | cksum | cut -c 1-6`
# random2=`head -20 /dev/urandom | cksum | cut -c 1-6`

# $NEMU_HOME/resource/simpoint/simpoint_repo/bin/simpoint   \
#     -loadFVFile $PROFILING_RES/${workload}/simpoint_bbv.gz \    # load a frequency vector file
#     -saveSimpoints $CLUSTER/simpoints0                     \    # file to save simpoints
#     -saveSimpointWeights $CLUSTER/weights0                 \    # file to save simpoints weights
#     -inputVectorsGzipped                                   \    # input vectors have been compressed with gzip
#     -maxK 3               \   # maximum number of clusters to use
#     -numInitSeeds 2       \   # times of different random initialization for each run, taking only the best clustering
#     -iters 1000           \   # maximum number of iterations that should perform
#     -seedkm ${random1}    \   # random seed for choosing initial k-means centers
#     -seedproj ${random2}  \   # random seed for random linear projection
#     > >(tee $log/${workload}-out.txt) 2> >(tee $log/${workload}-err.txt)
                                # redirect stdout and stderr
```

Checkpoint: SimPoint Checkpoint

```
------------------------------------------------------------
Run number 2 of at most 4, k = 3
------------------------------------------------------------
  ------------------------------------------------------------
  Initialization seed trial #1 of 2; initialization seed = 634853
  ------------------------------------------------------------
  Initialized k-means centers using random sampling: 3 centers
  Number of k-means iterations performed: 1
  BIC score: -27.9287
  Distortion: 7.02268
  Distortions/cluster: 0.542115 3.14007 3.34049
  Variance: 0.702268
  Variances/cluster: 0.271058 1.04669 0.668099
  ------------------------------------------------------------
  Initialization seed trial #2 of 2; initialization seed = 634854
  ------------------------------------------------------------
  Initialized k-means centers using random sampling: 3 centers
  Number of k-means iterations performed: 4
  BIC score: -10.3726
  Distortion: 6.04602
  Distortions/cluster: 5.29257 0.753145 0.000309052
  Variance: 0.604602
  Variances/cluster: 0.661571 0.753145 0.000309052
  The best initialization seed trial was #2
 ------------------------------------------------------------
 Post-processing runs
 ------------------------------------------------------------
 ------------------------------------------------------------
  For the BIC threshold, the best clustering was run 2 (k = 3)
```

Compute the information of K-means clustering

Obtain multiple program representative checkpoints

Checkpoint: SimPoint Checkpoint

- **Step 3: Generate corresponding Checkpoints based on clustering results**

```
$ bash simpoint_step3_genspt.sh        # It may take about 2 minutes
```

```
# simpoint_step3_genspt.sh (~2mins)

# export CLUSTER=$RESULT/cluster
# mkdir -p $LOG_PATH/checkpoint_logs

# $NEMU ${BBL_PATH}/${workload}.bin \
#     -b                           \  run with batch mode
#     -D $RESULT                   \  directory where the checkpoint is generated
#     -C checkpoint                \  name of this task
#     -w stream                    \  name of workload
#     -S $CLUSTER                  \  simpoints results director
#     --cpt-interval $interval     \  simpoint interval instructions: 50,000,000
#     > >(tee $log/stream-out.txt) 2> >(tee $log/stream-err.txt)
                                      redirect stdout and stderr
```

Checkpoint: SimPoint Checkpoint

```
[src/isa/riscv64/exec/special.c,38,exec_nemu_trap] Start profiling, resetting inst count

[src/checkpoint/serializer.cpp,100,serializeRegs] Writing int registers to checkpoint memory @[0x80001000, 0x80001100) [0x1000, 0x1100)
[src/checkpoint/serializer.cpp,110,serializeRegs] Writing float registers to checkpoint memory @[0x80001100, 0x80001200) [0x1100, 0x1200)
[src/checkpoint/serializer.cpp,118,serializeRegs] Writing PC: 0xffffffe000720f20 at addr 0x80001200
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x105: 0xffffffe000697cc8
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x106: 0xffffffffffffffff
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x141: 0x200013417e
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x142: 0x8
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x180: 0x80000000000f805d
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x300: 0xa00000022
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x301: 0x800000000014112d
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x302: 0xb109
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x303: 0x222
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x304: 0x22a
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x305: 0x800a0004
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x306: 0xffffffffffffffff
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x340: 0x800abdc0
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x341: 0xffffffe0008d6eb2
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x342: 0x2
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x3a0: 0x1f0800
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x3b0: 0x20028000
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x3b1: 0x2002d000
[src/checkpoint/serializer.cpp,141,serializeRegs] CSR 0x3b2: 0xffffffffffffffff
[src/checkpoint/serializer.cpp,144,serializeRegs] Writing CSR to checkpoint memory @[0x80001300, 0x80009300) [0x1300, 0x9300)
[src/checkpoint/serializer.cpp,152,serializeRegs] Touching Flag: 0xbeef at addr 0x80000f00
[src/checkpoint/serializer.cpp,156,serializeRegs] Record mode flag: 0x1 at addr 0x80000f08
[src/checkpoint/serializer.cpp,160,serializeRegs] Record time: 0x1 at addr 0x80000f10
[src/checkpoint/serializer.cpp,164,serializeRegs] Record time: 0x1 at addr 0x80000f18
[src/checkpoint/serializer.cpp,52,serializePMem] Created tutorial_simpoint/take_cpt/stream_0_0.083333/0/

Opening tutorial_simpoint/take_cpt/stream_0_0.083333/0/_0_.gz as checkpoint output file
[src/checkpoint/serializer.cpp,83,serializePMem] Written 0x7fffffff bytes

[src/checkpoint/serializer.cpp,83,serializePMem] Written 0x7fffffff bytes

[src/checkpoint/serializer.cpp,83,serializePMem] Written 0x7fffffff bytes

[src/checkpoint/serializer.cpp,83,serializePMem] Written 0x7fffffff bytes

[src/checkpoint/serializer.cpp,83,serializePMem] Written 0x4 bytes

[src/checkpoint/serializer.cpp,89,serializePMem] Checkpoint done!
```

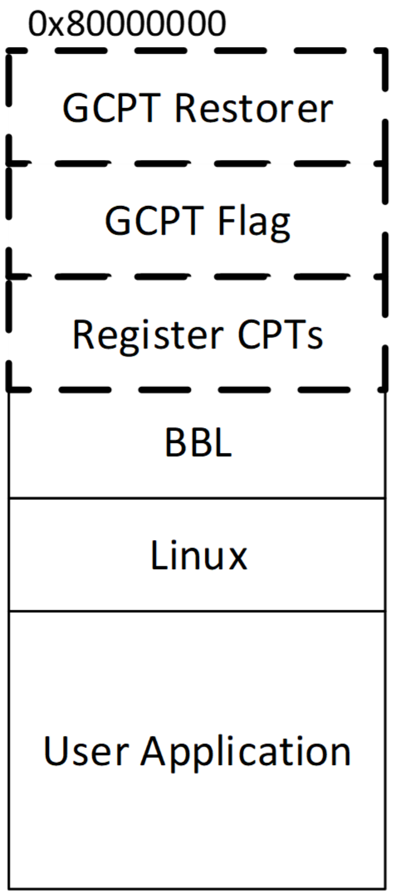
Start making checkpoint

Save int & float registers

Save CSR registers

Save other information

write checkpoints into .gz



Basic format of RISC-V Checkpoint

Checkpoint: SimPoint Checkpoint

- **Step 4: Run the SimPoint checkpoint on NEMU or XiangShan**
- Here we take XiangShan as an example

```
$ bash simpoint_step4_run_xs.sh
```

```
# simpoint_step4_run_xs.sh (~ 50s)

# ./emu \
#    -i `find $RESULT/checkpoint/stream -type f -name "*_.gz" | tail -1` \
                                    get the path of workload
#   --diff $NOOP_HOME/ready-to-run/riscv64-nemu-interpreter-so \
                                    Enable the reference standard design path for difftest
#   --max-cycles=50000 \            Maximum execution instruction count
#   2>simpoint.err
```

Checkpoint: SimPoint Checkpoint

- **Step 5: Dump configuration files for batch running on Gem5/XiangShan**

```
$ python3 simpoint_step5_dumpresult.py
$ ls -l simpoint_result/checkpoint
```

  - Collect profiling/checkpoint information to generate a list file for Gem5 execution

```
# checkpoint.lst
stream_5 stream/5 0 0 20 20
stream_11 stream/11 0 0 20 20
stream_1 stream/1 0 0 20 20
```

Workload Name | Checkpoint Path | Instructions to Skip (default 0) | Functional Warmup Instructions (default 0) | Detailed Warmup Instructions (default 20M) | Sample Instructions (default 20M)

  - Collect cluster information to generate a JSON file for XiangShan execution

```
# cluster.json
{"stream": {"insts": "654228721",
"points": {"7": "0.615385", "0": "0.153846", "11": "0.230769"}}}
```

Workload Name | Instruction Count | Checkpoint Index | Checkpoint Weight

Checkpoints: Uniform Checkpoint

- **Step 0: Prepare NEMU environment for checkpoints**

```
$ cd $XS_PROJECT_ROOT/tutorial/p4-checkpoint
$ bash simpoint_step0_prepare.sh
```

- **Step 1: Generate uniform checkpoints using NEMU**

```
$ bash uniform_cpt.sh
```

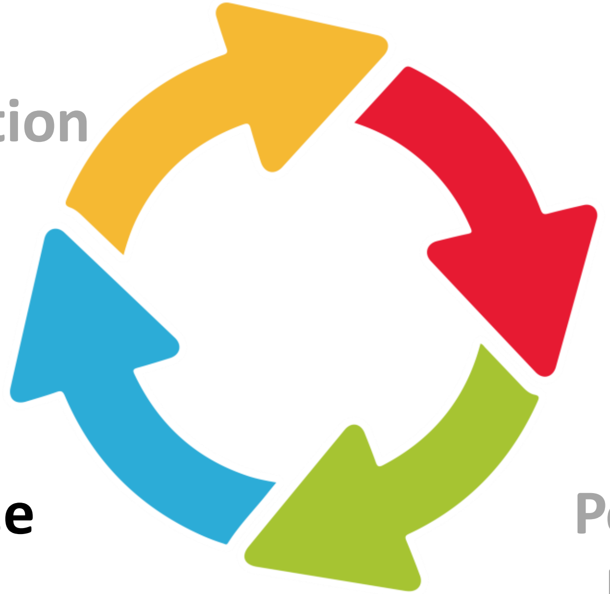
- **Step 2: Run uniform checkpoint on NEMU or XiangShan**
- Taking XiangShan as an example:

```
$ bash uniform_run_xs.sh
```

MinJie Performance Verification

Chisel-based prototypes

Implementation



Run Tests

gem5

VERILATOR

umd-memsys/
DRAMsim3

DRAMsim3: a Cycle-accurate, Thermal-Capable DRAM Simulator

Performance Evaluation

RISC-V Checkpoint

SimPoint

Performance Analysis

XSPerf

Top-Down

Constantin

XSPerf: Simulation Performance Collection

- **Purpose**
  - Collect simulation performance data under different requirements
- **XSPerf**
  - Multiple types for various fine-grained performance analysis
    - **Accumulation**: basic accumulator counter (log with stderr)
    - **Histogram**: count the distribution (log with stderr)
    - **Rolling**: rolling curve collection and visualization (ChiselDB)

XSPerf: Accumulate & Histogram

- **Example**

```
Accumulation:
[PERF][time=10000] ctrlBlock.rob: commitInstr,        1500
[PERF][time=10000] ctrlBlock.rob: waitLoadCycle,       800

Histogram:
[PERF][time=10000] l2cache: acquire_period_10_20,      15
[PERF][time=10000] l2cache: acquire_period_20_30,      200
[PERF][time=10000] l2cache: acquire_period_30_40,      60
```

XSPerf: Accumulate & Histogram

- **Usage**
  - Add *XSPerfAccumulate('name', signal)* or *XSPerfHistogram('name', signal)*
  - **By default**, it is printed after simulation finishes
  - Set *DebugOptions(EnablePerfDebug = false)* to turn off
- **Showcase**

```
$ cd ../p5-xs-perf
$ bash xs-perf-log.sh | head -n 20
```

```
# print the log result of p1-basic-func
# cat ${XS_PROJECT_ROOT}/tutorial/p1-basic-func/perf.err
```

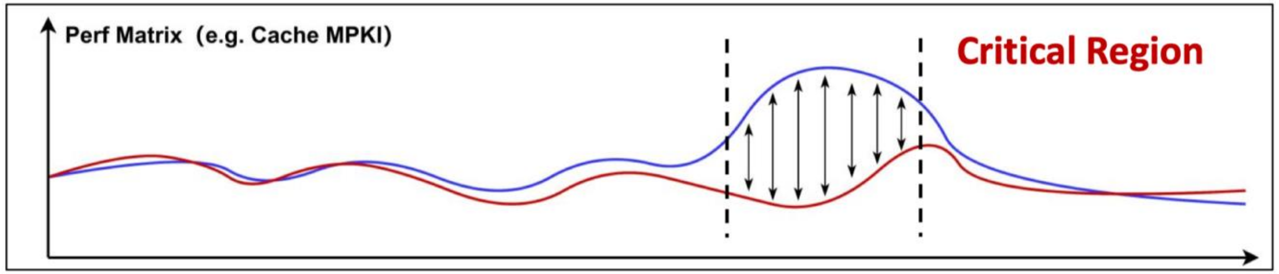
XSPerf: Accumulate & Histogram

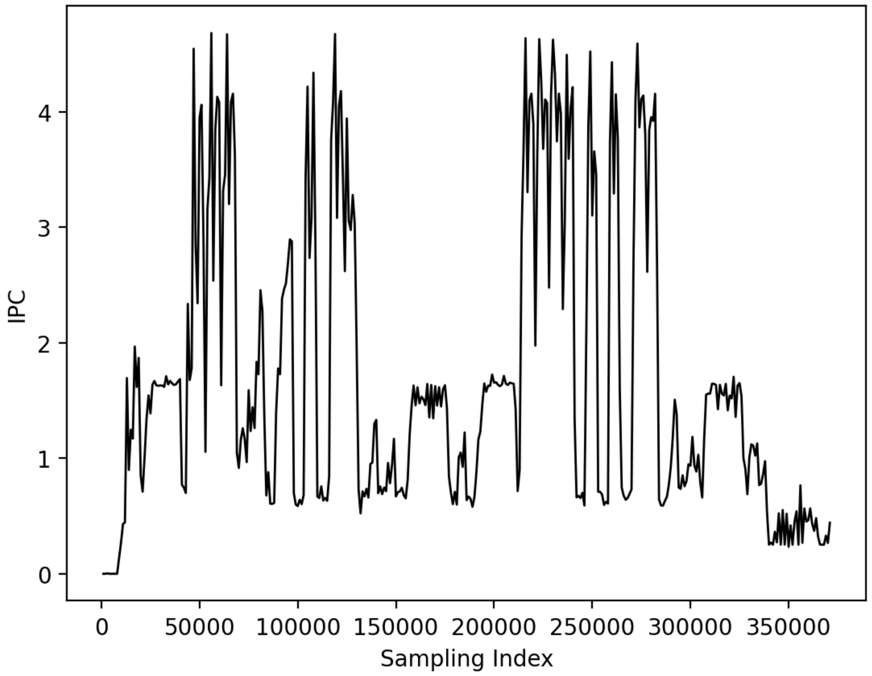
```
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.ctrlBlock.decode: utilization,          1183
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.bpu.predictors.components_3.tables_0.wrbypass: wrbypass_hit,          0
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.icache.missUnit.entries_1: entryPenalty1,          714
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.ctrlBlock.decode: waitInstr,          414
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.bpu.predictors.components_3.tables_0.wrbypass: wrbypass_miss,          0
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.ctrlBlock.decode: stall_cycle,          393
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.icache.missUnit.entries_1: entryReq1,          14
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.bpu.predictors.components_3.tables_1.wrbypass: wrbypass_hit,          0
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend: FrontendBubble,          4085
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.icache.missUnit.entries_0: entryPenalty0,          248
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.bpu.predictors.components_3.tables_1.wrbypass: wrbypass_miss,          2
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.ibuffer: utilization,          1826
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.icache.missUnit.entries_0: entryReq0,          6
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.ibuffer: util_0_1,          2029
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.ibuffer: util_1_2,          541
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.misc.busPMU: dcache_bank_0_A_channel_fire,          8
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.ibuffer: util_2_3,          45
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.misc.busPMU: dcache_bank_0_A_channel_stall,          0
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.ibuffer: util_3_4,          107
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.exuBlocks.scheduler.rs_4.staRS_0.oldestSelection: oldest_override_0,          0
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.misc.busPMU: dcache_bank_0_A_channel_PutFullData_fire,          0
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.ibuffer: util_4_5,          44
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.exuBlocks.scheduler.rs_4.staRS_0.oldestSelection: oldest_same_as_selected_0,          0
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.misc.busPMU: dcache_bank_0_A_channel_PutFullData_stall,          0
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.ibuffer: util_5_6,          85
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.exuBlocks.scheduler.rs_4.staRS_0.oldestSelection: oldest_override_1,          0
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.misc.busPMU: dcache_bank_0_A_channel_PutPartialData_fire,          0
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.ibuffer: util_6_7,          13
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.exuBlocks.scheduler.rs_4.staRS_0.oldestSelection: oldest_same_as_selected_1,          0
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.misc.busPMU: dcache_bank_0_A_channel_PutPartialData_stall,          0
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.ibuffer: util_7_8,          27
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.memBlock.dtlb_st_tlb_st.entries: port0_np_sp_multi_hit,          0
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.misc.busPMU: dcache_bank_0_A_channel_ArithmeticData_fire,          0
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.ibuffer: full,          862
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.memBlock.dtlb_st_tlb_st.entries: port1_np_sp_multi_hit,          0
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.misc.busPMU: dcache_bank_0_A_channel_ArithmeticData_stall,          0
[PERF ][time=          2893] TOP.SimTop.l_soc.core_with_l2.core.frontend.ibuffer: exHalf,          125
```

Find more parsing scripts at https://github.com/OpenXiangShan/env-scripts/blob/main/perf/perf.py

XSPerf: Rolling

- **Advantages**
  - Visualization of performance analysis
  - Detailed performance differences among segments
  - Help identify parameter-sensitive critical regions





XSPerf: Rolling

- **Usage**
  - Add *XSPerfRolling('name', perfCnt, granularity, clock, reset)* wherever you want
  - Compile with *WITH_CHISELDB=1 WITH_ROLLINGDB=1*, run with *--dump-db*
- **Showcase**

```
# Build emu with rolling  (time consuming, use pre-built emu instead)
# bash xs-perf-prepare.sh
```

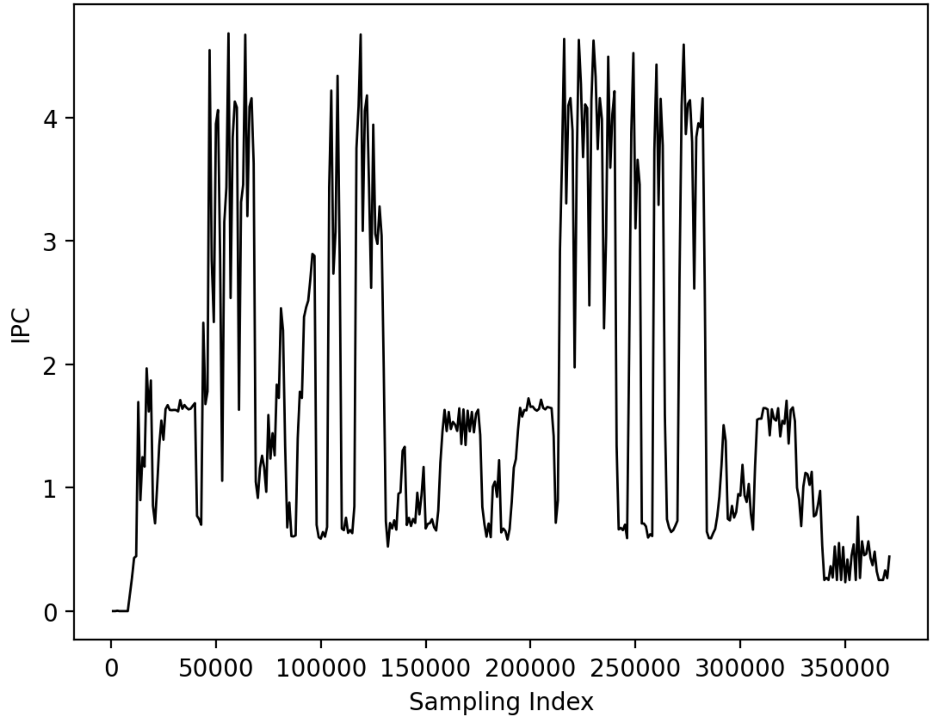
```
# cd ${NOOP_HOME}
# make clean
# make emu EMU_THREADS=4 WITH_CHISELDB=1 WITH_ROLLINGDB=1 -j8 \
#     PGO_WORKLOAD=${NOOP_HOME}/ready-to-run/coremark-2-iteration.bin \
#     PGO_MAX_CYCLE=10000 PGO_EMU_ARGS=--no-diff LLVM_PROFDATA=llvm-profdata
# ./build/emu -i ./ready-to-run/coremark-2-iteration.bin \
#     --diff ./ready-to-run/riscv64-nemu-interpreter-so --dump-db
# cp `find $NOOP_HOME/build/ -type f -name "*.db" | tail -1` \
#     ${XS_PROJECT_ROOT}/tutorial/p5-xs-perf/xs-perf-rolling.db
```

XSPerf: Rolling

- **Hands-on:** check the results and plot the curve of performance segments

```
# tutorial/p5-xs-perf/
$ bash xs-perf-rolling.sh
```

```
# cd ${NOOP_HOME}/scripts/rolling
# python3 rollingplot.py
${XS_PROJECT_ROOT}/tutorial/p6-xs-
perf/XsPerfRolling-test.db ipc
# ls ${NOOP_HOME}/scripts/rolling/results

perf.png
```



rolling curve example

Top-Down: Method for Performance Analysis

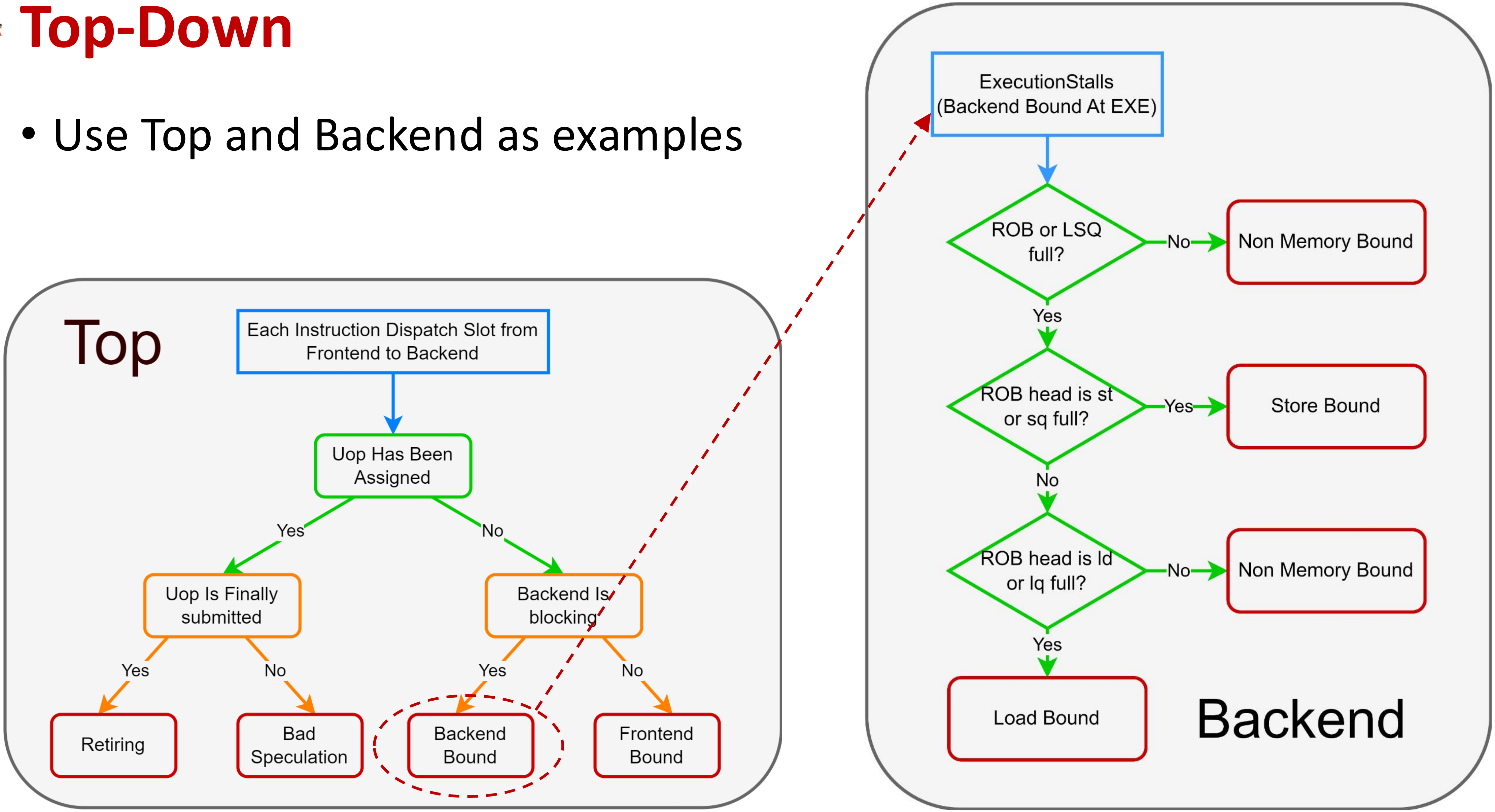
- **Purpose**
  - Organize scattered performance events in a **hierarchical manner**
  - **Accurately** calculate one event's impact on processor performance

- We apply the **Top-Down** approach to XS-Gem5 and XiangShan
  - Complete **targeted optimization adaptation** for RISC-V instruction set
  - Optimize **performance counters** according to XiangShan microarchitecture
  - Further refine the **hierarchical design** of Top-Down model
  - Without missing or duplicating any events
  - Without assuming the blocking cycles of performance events in advance

Yasin, Ahmad. "A Top-Down Method for Performance Analysis and Counters Architecture." 2014 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), 2014, pp. 35–44, https://doi.org/10.1109/ISPASS.2014.6844459.

Top-Down

- Use Top and Backend as examples



Top-Down in RTL

- Setting Performance Counters in RTL Code

```
XiangShan/src/main/scala/xiangshan/backend/dispatch/Dispatch.scala

val stallReason = Wire(chiselTypeOf(io.stallReason.reason))
// ...
TopDownCounters.values.foreach(ctr =>
  XSPerfAccumulate(ctr.toString(), PopCount(stallReason.map(_ === ctr.id.U)))
)
```

- Do simulation to collect performance counter data

Top-Down in RTL

- Analyze the performance counter data through Top-Down method

```
XiangShan/scripts/top-down/configs.py


xs_coarse_rename_map = {
  'OverrideBubble': 'MergeFrontend',
  'FtqFullStall': 'MergeFrontend',
  'IntDqStall': 'MergeCoreDQStall',  # attribute perf-counters to Top-Down hierarchy
  'FpDqStall': 'MergeCoreDQStall’
}
```

- Obtain analysis results and make targeted optimizations

Top-Down in RTL

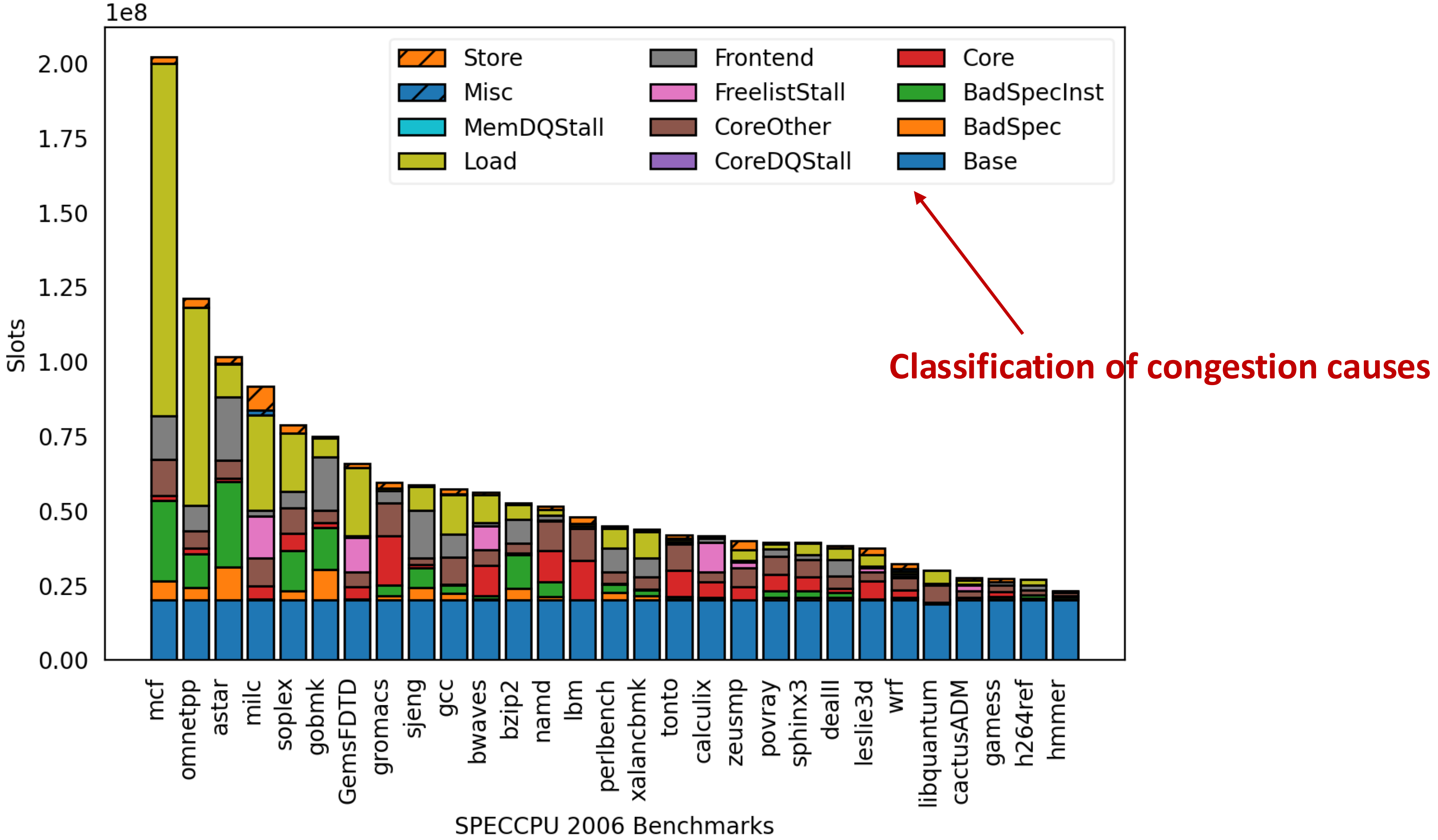
- **Hands-on:** analyze the congestion causes in RTL using the Top-Down tool

```
#run the Top-Down analysis scripts (it takes ~1min)

$ bash rtl-top-down.sh
```

```
# cd ${NOOP_HOME}/scripts/top-down && \
  python3 top_down.py \
  -s /opt/SPEC06_EmuTasks_topdown \     #path of performance counter results
  -j /opt/SPEC06_EmuTasks_topdown.json #json file of checkpoints configuration

# ls ${NOOP_HOME}/scripts/top-down/results
result.png  results.csv  results-weighted.csv #output of visual analysis results
```

Top-Down in RTL

- Result Example: Top-Down visual analysis results



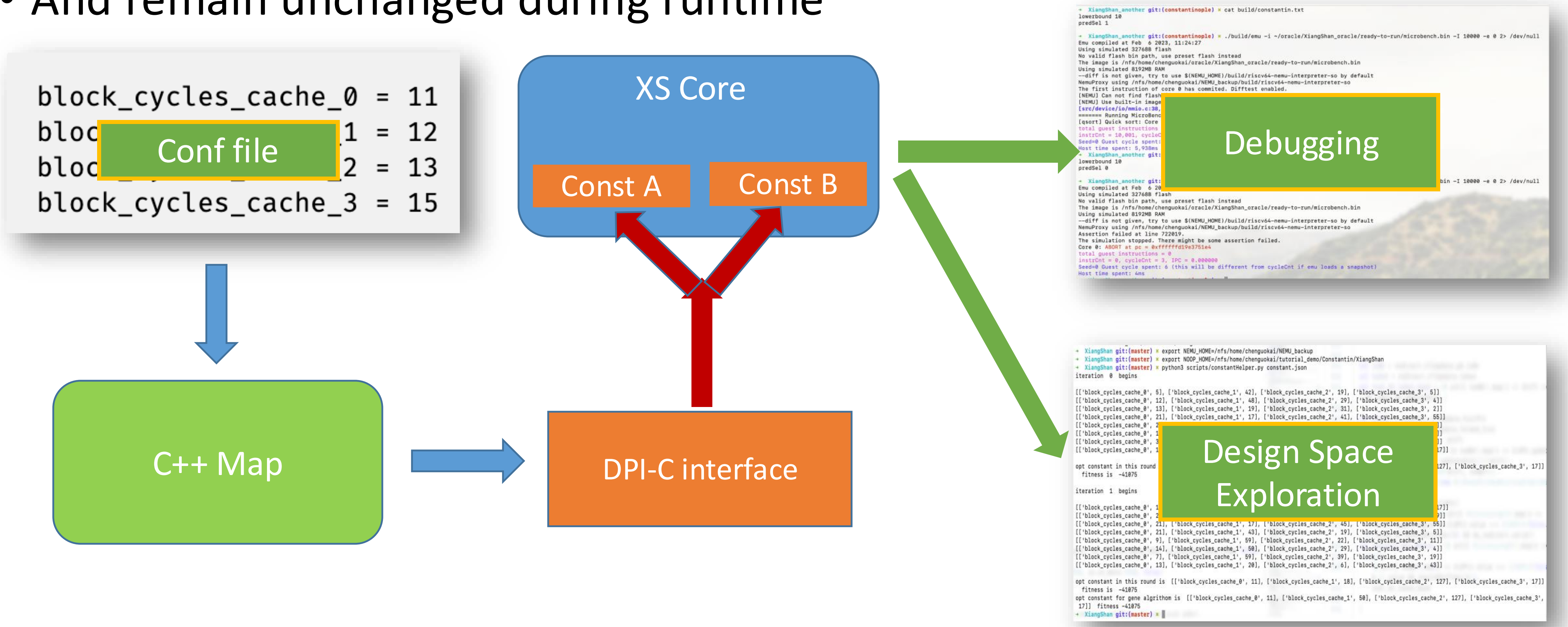
ConstantIn

- **Motivation**
  - Want to test the performance under different parameters
  - But re-compilation is time consuming
  - Can we change parameters (constants) at runtime?

- **ConstantIn**
  - DPI-C: Using C++ function in Chisel code (through Verilog using BlackBox)
  - Signal values configured at runtime rather than compile time

ConstantIn

- Constants can be changed when initializing
- And remain unchanged during runtime



ConstantIn

- **Usage:** Create signals

```
# XiangShan/coupledL2/src/main/scala/coupledL2/prefetch/TemporalPrefetch.scala
```

```scala
require(cacheParams.hartIds.size == 1)
val hartid = cacheParams.hartIds.head
// 0 / 1: whether to enable temporal prefetcher
private val enableTP = WireInit(Constantin.createRecord("enableTP"+hartid.toString, initValue = 1.U))
// 0 ~ N: throttle cycles for each prefetch request
private val tpThrottleCycles = WireInit(Constantin.createRecord("tp_throttleCycles"+hartid.toString, initValue = 4.U(3.W)))
// 0 / 1: whether request to set as trigger on meta hit
private val hitAsTrigger = WireInit(Constantin.createRecord("tp_hitAsTrigger"+hartid.toString, initValue = 1.U))
// 1 ~ triggerQueueDepth: enqueue threshold for triggerQueue
private val triggerThres = WireInit(Constantin.createRecord("tp_triggerThres"+hartid.toString, initValue = 1.U(3.W)))
// 1 ~ tpEntryMaxLen: record threshold for recorder and sender (storage size will not be affected)
private val recordThres = WireInit(Constantin.createRecord("tp_recordThres"+hartid.toString, initValue = tpEntryMaxLen.U))
// 0 / 1: whether to train on vaddr
private val trainOnVaddr = WireInit(Constantin.createRecord("tp_trainOnVaddr"+hartid.toString, initValue = 0.U))
// 0 / 1: whether to eliminate L1 prefetch request training
private val trainOnL1PF = WireInit(Constantin.createRecord("tp_trainOnL1PF"+hartid.toString, initValue = 0.U))
```

ConstantIn

- **Hands-on:** Pass constant via standard input stream
  - **By default**, read constant via *${NOOP_HOME}/build/constantin.txt*
  - **For better demonstration**, it is changed to stdin (corresponding modifications can refer to p6-constantin/*utility.patch*)

```
$ cd ../p6-constantin
```

```
# bash step0-build.sh     # Build emu with constantin (time consuming, use pre-built emu instead)
```

```
$ bash step1-basic.sh     # Run emu and pass constant
please input total constant number
2
please input each constant ([constant name] [value])
pbop_delayQueueLatency 150
vbop_delayQueueLatency 150
```

ConstantIn

- **Feature:** *AutoSolving* for automatically finding better constant values
- **Process**
  - Enable *AutoSolving* & prepare signals you need
  - Compile the emu with *WITH_CONSTANTIN=1*
  - Set parameters for *AutoSolving* in a configuration file
  - Run the script to find automatically

ConstantIn

- **Feature:** *AutoSolving* for automatically finding better constant values

```
# Automatical parameter solver configuration file

$ cat my_constantin.json
```


```
{
    "constants": [
        {
            "name": "vbop_delayQueueLatency",
            "width": 8,
            "guide": 256,
            "init": 177
        },
        {
            "name": "pbop_delayQueueLatency",
            "width": 8,
            "guide": 256,
            "init": 242
        }
    ],
    "opt_target": [
        {"l2cache.topDown: L2MissMatch,": {"policy" :"min", "baseline" :409} },
        {"backend.ctrlBlock.rob: clock_cycle,": {"policy" :"min", "baseline" :217759} }
    ],

    "population_num": 4,
    "iteration_num": 2,
    "crossover_rate": 50,
    "mutation_rate": 50,

    "emu_threads": 8,
    "concurrent_emu": 4,
    "max_instr": 50000,
    "seed": 3888,
    "work_load": "${XS_PROJECT_ROOT}/tutorial/p6-constantin/maprobe-riscv64-xs.bin"
}
```


- **Parameters defined**
  - properties of signals
  - target of *AutoSolving*
  - parameters of genetic algorithm
  - parameters of XS emulator

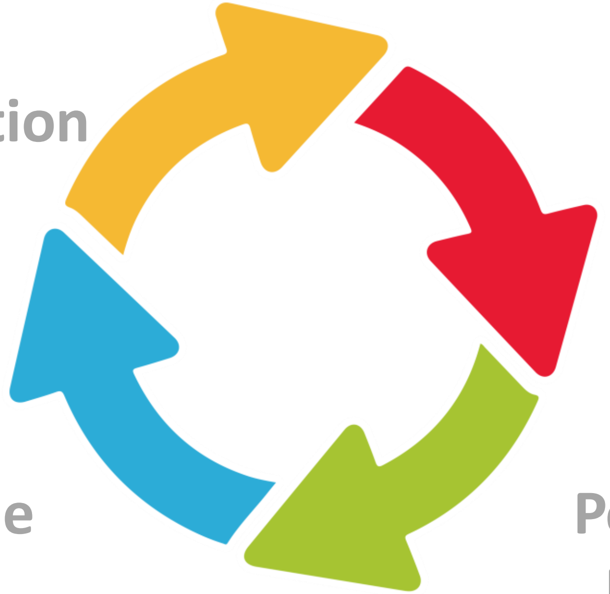
ConstantIn

- **Hands-on:** *AutoSolving* example

```
# Run auto solver, output will be the optimal constant currently found
$ bash step2-solve.sh
```

```
#  ~2min
# The solver generates optimal parameters
opt constant in this round is  [['vbop_delayQueueLatency', 242], ['pbop_delayQueueLatency', 135]]  fitness is  -316
opt constant for gene algrithom is  [['vbop_delayQueueLatency', 243], ['pbop_delayQueueLatency', 135]]  fitness -316
```

MinJie Performance Verification



Chisel-based prototypes

Implementation

Run Tests

gem5

VERILATOR

umd-memsys/
DRAMsim3

DRAMsim3: a Cycle-accurate, Thermal-Capable DRAM Simulator

XSPerf

Constantin

Top-Down

Performance Analysis

Performance Evaluation

RISC-V Checkpoint

SimPoint

XS-Gem5: Architecture Performance Evaluation Tool

- **Purpose**
  - Explore microarchitecture designs efficiently
  - Conduct fast end-to-end performance evaluations
- **XS-Gem5**
  - Align with XiangShan RTL based on Gem5 O3CPU with modifications
  - Support Difftest and Checkpoint
  - Scripts for XiangShan performance analysis

XS-Gem5

- **Hands-on:** compile and build XS-Gem5

```
$ cd ../p7-xs-gem5 # Enter XS-Gem5 tutorial directory
$ ~~bash 0-gem5_prepare.sh~~ # We have built this for you, skip it on our server to save cpu time and memory
```

```
prepare_gem5() {
    pushd $gem5_home && \
    cd ext/dramsim3 && \ # build DRAMSim3
    git clone git@github.com:umd-memsys/DRAMSim3.git DRAMsim3 && \
    cd DRAMsim3 && mkdir -p build && cd build && cmake .. && make -j `nproc` && \
    popd
}
build_gem5() {
    pushd $gem5_home && \
    scons build/RISCV/gem5.opt --gold-linker -j `nproc` && \
    popd
}
```

XS-Gem5

- **Hands-on:** run bare-metal workload on XS-Gem5

```
$ bash 1-gem5_run_coremark.sh
```

```
#~9s
pushd $gem5_home && \
export GCBV_REF_SO=$gem5_home/ext/NEMU/build/riscv64-nemu-interpreter-so && \
mkdir -p util/xs_scripts/coremark && \ # Setup CoreMark working directory
cd util/xs_scripts/coremark && \
$gem5_home/build/RISCV/gem5.opt $gem5_home/configs/example/xiangshan.py \
--raw-cpt \
--generic-rv-cpt=$NOOP_HOME/ready-to-run/coremark-2-iteration.bin && \
popd
```

XS-Gem5

```
build/RISCV/mem/physical.cc:608: info: First 4 bytes are 0x13 0x4 0x0 0x0
build/RISCV/sim/system.cc:561: info: Restored from Xiangshan RISC-V Checkpoint
**** REAL SIMULATION ****
build/RISCV/sim/simulate.cc:194: info: Entering event queue @ 0.  Starting simulation...
build/RISCV/cpu/base.cc:1266: warn: Start memcpy to NEMU from 0x7eb525400000, size=8589934592
build/RISCV/cpu/base.cc:1269: warn: Start regcpy to NEMU
build/RISCV/dev/serial/uartlite.cc:35: warn: Write to other uartlite addr 12 is not implemented
Running CoreMark for 2 iterations
2K performance run parameters for coremark.
CoreMark Size     : 666
Total time (ms)   : 110
Iterations        : 2
Compiler version  : GCC10.2.0
seedcrc           : 0xe9f5
[0]crclist        : 0xe714
[0]crcmatrix      : 0x1fd7
[0]crcstate       : 0x8e3a
[0]crcfinal       : 0x72be
Finised in 110 ms.
==================================================
CoreMark Iterations/Sec 18181
Exiting @ tick 313394625 because m5_exit instruction encountered when simulating XS
```

XS-Gem5

- **Hands-on:** analyze the performance counters

```
$ bash 2-gem5_counter.sh | head -n 20
```

```
cat $gem5_home/util/xs_scripts/coremark/m5out/stats.txt
```

```
---------- Begin Simulation Statistics ----------
simSeconds                                   0.000313                       # Number of seconds simulated (Second)
simTicks                                    313394625                       # Number of ticks simulated (Tick)
finalTick                                   313394625                       # Number of ticks from beginning of simulation (restored from checkpoints and never reset) (Tick)
simFreq                                 1000000000000                       # The number of ticks per simulated second ((Tick/Second))
hostSeconds                                      8.53                       # Real time elapsed on the host (Second)
hostTickRate                                 36722157                       # The number of ticks simulated per host second (ticks/s) ((Tick/Second))
hostMemory                                   17143924                       # Number of bytes of host memory used (Byte)
simInsts                                       626869                       # Number of instructions simulated (Count)
simOps                                         626869                       # Number of ops (including micro ops) simulated (Count)
hostInstRate                                    73453                       # Simulator instruction rate (inst/s) ((Count/Second))
hostOpRate                                      73453                       # Simulator op (including micro ops) rate (op/s) ((Count/Second))
system.bridge.power_state.pwrStateResidencyTicks::UNDEFINED    313394625                       # Cumulative time (in ticks) in various power states (Tick)
system.clk_domain.clock                           333                       # Clock period in ticks (Tick)
system.cpu.numCycles                           941131                       # Number of cpu cycles simulated (Cycle)
system.cpu.numWorkItemsStarted                      0                       # Number of work items this cpu started (Count)
system.cpu.numWorkItemsCompleted                    0                       # Number of work items this cpu completed (Count)
system.cpu.instsAdded                          784076                       # Number of instructions added to the IQ (excludes non-spec) (Count)
system.cpu.nonSpecInstsAdded                        3                       # Number of non-speculative instructions added to the IQ (Count)
```

XS-Gem5: Performance Analysis Scripts

- **Hands-on:** run the Top-Down analysis script

```
$ bash 3-gem_topdown.sh
```

```
pushd gem5_data_proc && \
python3 batch.py \
  -s $gem5_home/util/xs_scripts/coremark \
  -t --topdown-raw && \
popd
```

XS-Gem5: Performance Analysis Scripts - TopDown

```
~/pre/tutorial/p7-xs-gem5/gem5_data_proc ~/pre/tutorial/p7-xs-gem5
workload: m5out, point: 0, segments: 1
[('m5out_0', '/home/ecs-user/gem5-pre/gem5/util/xs_scripts/coremark/m5out/stats.txt')]
Process finished job: m5out_0
/home/ecs-user/gem5-pre/gem5/util/xs_scripts/coremark/m5out/stats.txt
{'m5out_0': {'OtherFetchStall': 0, 'OtherStall': 0, 'CommitSquash': 163020, 'ResumeUnblock': 0
, 'OtherMemStall': 60, 'Atomic': 0, 'MemCommitRateLimit': 2382, 'MemNotReady': 1718802, 'MemSq
uashed': 0, 'StoreMemBound': 0, 'StoreL3Bound': 0, 'StoreL2Bound': 0, 'StoreL1Bound': 2232, 'L
oadMemBound': 4470, 'LoadL3Bound': 0, 'LoadL2Bound': 0, 'LoadL1Bound': 672, 'InstNotReady': 20
4, 'SerializeStall': 63792, 'InstSquashed': 44832, 'InstMisPred': 0, 'FetchBufferInvalid': 0,
'SquashStall': 7290, 'FragStall': 1715806, 'TrapStall': 0, 'IntStall': 0, 'BpStall': 116700, '
DTlbStall': 192, 'ITlbStall': 0, 'IcacheStall': 224154, 'NoStall': 778901, 'ipc': 0.758999, 'I
nsts': 663614, 'Cycles': 874328, 'point': '0', 'workload': 'm5out', 'bmk': 'm5out'}}
         Atomic  BpStall  CommitSquash  Cycles  ...   bmk    ipc  point  workload
m5out_0       0   116700        163020  874328  ...  m5out  0.759      0     m5out

[1 rows x 37 columns]
```

XS-Gem5: Performance Analysis Scripts – Cache MPKI

- **Hands-on:** run the Cache MPKI analysis script

```
$ bash 4-gem5_cache.sh
```

```
pushd gem5_data_proc && \
mkdir -p results && \
export PYTHONPATH=`pwd` && \
python3 batch.py \
  -s /home/share/xs-model-l1bank \
  -o gem5-cache-example.csv \
  --cache && \
python3 simpoint_cpt/compute_weighted.py \
  -r gem5-cache-example.csv \
  -j /home/share/xs-model-l1bank/cluster-0-0.json \
  -o weighted.csv
```

Note: Here we take the results from existing output from running SPEC CPU 2006 checkpoints.

XS-Gem5: Performance Analysis Scripts – Cache MPKI

```
['GemsFDTD', 'astar', 'bwaves', 'bzip2', 'cactusADM', 'calculix', 'dealII', 'gamess', 'gcc', 'gobmk', 'gromacs', 'h264ref', 'hmmer', 'lbm', 'leslie3d', 'libquantum', 'mcf',
 'milc', 'namd', 'omnetpp', 'perlbench', 'povray', 'sjeng', 'soplex', 'sphinx3', 'tonto', 'wrf', 'xalancbmk', 'zeusmp']
             Cycles      Insts  dcache_miss  l1d.data.MPKI  l1d.data.miss  l2.data.MPKI  ...     l3_acc  l3_acc_l1_pref     l3_miss  l3_miss_l1d_pref    cpi  coverage
perlbench  8.118e+06  2.000e+07    1.068e+05          5.338      1.068e+05           0.0  ...  4.159e+04       3.076e+04   23625.296         18072.642  0.406     0.820
bzip2      9.541e+06  2.000e+07    2.442e+05         12.212      2.442e+05           0.0  ...  9.817e+04       5.423e+04    4635.961          3321.498  0.477     0.988
gcc        9.261e+06  2.000e+07    2.785e+05         13.926      2.785e+05           0.0  ...  4.285e+05       3.880e+05  114394.979        104616.710  0.463     1.000
mcf        3.027e+07  2.000e+07    5.413e+06        270.673      5.413e+06           0.0  ...  2.294e+06       1.575e+06  698680.709        575668.623  1.513     1.000
gobmk      1.155e+07  2.000e+07    1.097e+05          5.485      1.097e+05           0.0  ...  3.856e+04       3.230e+04   14055.195         10964.674  0.577     1.000
hmmer      4.489e+06  2.000e+07    2.084e+04          1.042      2.084e+04           0.0  ...  2.653e+04       2.627e+04     494.690           418.687  0.224     1.000
sjeng      9.021e+06  2.000e+07    2.879e+04          1.439      2.879e+04           0.0  ...  2.422e+04       1.448e+04   20958.788         13269.176  0.451     1.000
libquantum 4.390e+06  2.000e+07    2.371e+04          1.185      2.371e+04           0.0  ...  5.120e+05       5.110e+05   69486.561         68474.121  0.220     1.000
h264ref    4.941e+06  2.000e+07    7.420e+04          3.710      7.420e+04           0.0  ...  3.053e+04       2.751e+04   10893.001          9186.843  0.247     1.000
omnetpp    1.774e+07  2.000e+07    1.154e+06         57.713      1.154e+06           0.0  ...  1.026e+06       7.398e+05  492397.438        413052.400  0.887     1.000
astar      1.618e+07  2.000e+07    5.450e+05         27.252      5.450e+05           0.0  ...  3.543e+05       2.588e+05   48744.997         41617.766  0.809     1.000
xalancbmk  6.515e+06  2.000e+07    3.998e+05         19.989      3.998e+05           0.0  ...  3.246e+05       2.924e+05   38865.836         30536.527  0.326     1.000
bwaves     8.071e+06  2.000e+07    1.296e+05          6.480      1.296e+05           0.0  ...  5.694e+05       5.677e+05   89426.320         87864.638  0.404     1.000
gamess     4.261e+06  2.000e+07    4.574e+04          2.287      4.574e+04           0.0  ...  3.496e+03       3.286e+03    2324.679          2228.684  0.213     1.000
milc       1.071e+07  2.000e+07    1.375e+06         68.772      1.375e+06           0.0  ...  9.773e+05       9.581e+05  702851.431        692453.110  0.535     1.000
zeusmp     7.450e+06  2.000e+07    2.486e+05         12.428      2.486e+05           0.0  ...  1.894e+05       1.805e+05   85596.788         82176.235  0.373     1.000
gromacs    9.386e+06  2.000e+07    6.421e+05         32.104      6.421e+05           0.0  ...  1.976e+04       1.865e+04    7220.444          6523.288  0.469     1.000
cactusADM  4.474e+06  2.000e+07    4.308e+04          2.154      4.308e+04           0.0  ...  9.884e+04       9.758e+04   34987.191         34295.763  0.224     1.000
leslie3d   6.552e+06  2.000e+07    4.255e+05         21.274      4.255e+05           0.0  ...  3.842e+05       3.793e+05  144669.463        142838.003  0.328     1.000
namd       6.494e+06  2.000e+07    2.828e+05         14.142      2.828e+05           0.0  ...  5.496e+03       5.306e+03    4312.841          4150.029  0.325     1.000
dealII     5.946e+06  2.000e+07    1.630e+05          8.148      1.630e+05           0.0  ...  8.997e+04       8.439e+04   10109.797          7289.549  0.297     1.000
soplex     1.119e+07  2.000e+07    6.678e+05         33.391      6.678e+05           0.0  ...  9.042e+05       7.884e+05  175472.264        155213.001  0.559     1.000
povray     6.162e+06  2.000e+07    3.497e+05         17.487      3.497e+05           0.0  ...  3.944e+03       3.765e+03    3190.261          3044.166  0.308     1.000
calculix   6.720e+06  2.000e+07    1.863e+04          0.932      1.863e+04           0.0  ...  8.947e+03       8.803e+03    3756.312          3651.163  0.336     1.000
GemsFDTD   6.559e+06  2.000e+07    2.564e+05         12.818      2.564e+05           0.0  ...  5.250e+05       5.049e+05  374770.295        359822.201  0.328     1.000
tonto      6.759e+06  2.000e+07    1.067e+05          5.334      1.067e+05           0.0  ...  8.166e+04       7.943e+04    5793.963          5075.233  0.338     1.000
lbm        7.920e+06  2.000e+07    1.606e+05          8.029      1.606e+05           0.0  ...  6.969e+05       6.896e+05  122392.254        119831.519  0.396     1.000
wrf        4.908e+06  2.000e+07    5.679e+04          2.839      5.679e+04           0.0  ...  1.151e+05       1.133e+05   12708.207         12142.350  0.245     1.000
sphinx3    5.736e+06  2.000e+07    2.923e+05         14.614      2.923e+05           0.0  ...  3.751e+05       3.447e+05   20540.266         19216.285  0.287     1.000

[29 rows x 17 columns]
```

XS-Gem5: Performance Analysis Scripts

- **Hands-on:** run the SPEC CPU score calculation script

```
$ bash 5-gem5_spec06_score.sh
```

```
pushd gem5_data_proc && \
mkdir -p results && \
export PYTHONPATH=`pwd` && \
python3 batch.py -s ../data/xs-model-l1bank \
  -o gem5-score-example.csv && \
python3 simpoint_cpt/compute_weighted.py \
  -r gem5-score-example.csv \
  -j ../data/xs-model-l1bank/cluster-0-0.json \
  --score score.csv
```

Note: Here we take the results from existing output from running SPEC CPU 2006 checkpoints.

SPEC CPU score

```
wrf          250.853   11170.0   14.843          [25/2955]
leslie3d     214.233    9400.0   14.626          1.0
gamess       451.597   19580.0   14.452          1.0
namd         210.250    8020.0   12.715          1.0
tonto        260.012    9840.0   12.615          1.0
hmmer        260.274    9330.0   11.949          1.0
gromacs      201.133    7140.0   11.833          1.0
perlbench    289.646    9770.0   11.244          1.0
sjeng        366.934   12100.0   10.992          1.0
gobmk        320.031   10490.0   10.926          1.0
astar        229.743    7020.0   10.185          1.0
bzip2        410.138    9650.0    7.843          1.0
calculix     490.428    8250.0    5.607          1.0
score.csv
================ SPEC06 =================
================ Int ================
             time     ref_time    score   coverage
perlbench    289.646    9770.0   11.244        1.0
bzip2        410.138    9650.0    7.843        1.0
gcc          167.812    8050.0   15.990        1.0
mcf          141.107    9120.0   21.544        1.0
gobmk        320.031   10490.0   10.926        1.0
hmmer        260.274    9330.0   11.949        1.0
sjeng        366.934   12100.0   10.992        1.0
libquantum   148.529   20720.0   46.500        1.0
h264ref      428.160   22130.0   17.229        1.0
omnetpp      139.743    6250.0   14.908        1.0
astar        229.743    7020.0   10.185        1.0
xalancbmk     85.528    6900.0   26.892        1.0
Estimated Int score per GHz: 15.118648674723131
Estimated Int score @ 3.0GHz: 45.35594602416939
            time     ref_time    score   coverage
bwaves     175.907   13590.0   25.752        1.0
gamess     451.597   19580.0   14.452        1.0
milc       142.358    9180.0   21.495        1.0
```

```
mcf          141.107    9120.0   21.544          1[5/2983
gobmk        320.031   10490.0   10.926          1.0
hmmer        260.274    9330.0   11.949          1.0
sjeng        366.934   12100.0   10.992          1.0
libquantum   148.529   20720.0   46.500          1.0
h264ref      428.160   22130.0   17.229          1.0
omnetpp      139.743    6250.0   14.908          1.0
astar        229.743    7020.0   10.185          1.0
xalancbmk     85.528    6900.0   26.892          1.0
Estimated Int score per GHz: 15.118648674723131
Estimated Int score @ 3.0GHz: 45.35594602416939
================ FP =================
              time     ref_time    score   coverage
bwaves      175.907   13590.0   25.752        1.0
gamess      451.597   19580.0   14.452        1.0
milc        142.358    9180.0   21.495        1.0
zeusmp      196.183    9100.0   15.462        1.0
gromacs     201.133    7140.0   11.833        1.0
cactusADM   252.748   11950.0   15.760        1.0
leslie3d    214.233    9400.0   14.626        1.0
namd        210.250    8020.0   12.715        1.0
dealII      144.375   11440.0   26.413        1.0
soplex      136.694    8340.0   20.337        1.0
povray       93.311    5320.0   19.005        1.0
calculix    490.428    8250.0    5.607        1.0
GemsFDTD    176.662   10610.0   20.019        1.0
tonto       260.012    9840.0   12.615        1.0
lbm         149.280   13740.0   30.681        1.0
wrf         250.853   11170.0   14.843        1.0
Estimated FP score per GHz: 16.548846674376477
Estimated FP score @ 3.0GHz: 49.64654002312943
================ overall ================
Estimated overall score per GHz: 15.941323843383495
Estimated overall score @ 3.0GHz: 47.82397153015049
~/xs-workspace/xs-env-tutorial/tutorial/p7-xs-gem5
```

Summary: MinJie Development Flows and Tools

New feature Documentation → **RTL Implementation** → **Functional Verification** → **Performance Verification** → **Physical Design**

RTL Implementation
- Basic Architecture
- Design language

Functional Verification
- Test generation Nexus AM
- Reference model NEMU
- Result comparison Difftest
- Error reproducing LightSSS
- Error analysis ChiselDB, TL-Test

Performance Verification
- Test generation checkpoints
- Exploration model XS-Gem5
- Perf. Collection XSPerf
- Bottleneck analysis Top-Down
- Dynamic tuning ConstantIn

Thanks!

This tutorial is organized by Haoyuan Feng, YueXiao Ma, Zehao Liu, Zhida Lin, Yangyu Chen.